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STATE-OF-THE-ART PAPER

Transit network design problem: a half century of methodological research

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Abstract

This study presents the most extensive and temporally grounded review of the transit network design problem (TNDP) to date, covering five decades of research and offering a unified perspective on its two core subcomponents: the transit route network design problem (TRNDP) and the frequency setting problem (FSP). As cities face mounting pressures from urbanization, climate change, and equity demands, the strategic design of public transit networks has become increasingly critical. Despite the problem's centrality to transportation planning, the field remains fragmented and methodologically saturated, lacking integrated approaches that reflect real-world complexity. This review addresses that gap by analyzing over 170 studies from 1970 to 2024, systematically categorizing them by methodology, objective function, network scale, and system application. It is the first study to employ decade-resolved visual analytics, including heatmaps and taxonomies, to illustrate methodological trends, such as the rise of metaheuristics in the 2010s, the emerging—but still limited—role of AI/ML post-2020, and the declining prominence of classical optimization models. The study also introduces a novel scalability–performance matrix, comparing 10 solution approaches across multiple dimensions, and highlights the integration of TRNDP and FSP as a pivotal frontier in transit research. In doing so, it reveals critical research gaps, particularly the lack of resilience, equity, and adaptability in existing models, and proposes a forward-looking agenda rooted in unified, real-time, and data-driven frameworks. The review offers both a historical map and a strategic roadmap for scholars and practitioners seeking to advance sustainable and inclusive urban transit systems. The scientific value of this work lies in its combination of historical depth and methodological synthesis, introducing a novel scalability–performance matrix, decade-resolved visual analytics, and an integration-focused framework that has not been attempted in prior reviews.

Keywords Transit networks · Route design · Frequency optimization · Metaheuristics · Artificial Intelligence

Introduction

Efficient and equitable public transportation systems are essential for addressing the growing challenges of urbanization, including traffic congestion, environmental degradation, and social inequality [69]. Among the various planning activities that underpin transit systems, transit network design stands out as the strategic foundation upon which all operational and tactical decisions are built. The process of transit planning typically involves five sequential stages: network design, frequency setting, timetable development, vehicle scheduling, and crew scheduling. Of these, the transit network design problem (TNDP) plays a central role, as the structure of the transit network directly influences all subsequent planning stages and determines the accessibility, cost-efficiency, and usability of the system [137, 138].

The TNDP focuses on determining an optimal set of transit routes and frequencies that balance the often conflicting interests of stakeholders, primarily transit operators and users. Operator-related goals typically focus on minimizing operational costs and fleet size, while user-related objectives aim to reduce travel times, the number of transfers, and spatial inequities in service distribution. These objectives are frequently in tension, making the TNDP inherently a multi-objective, combinatorial, and NP-hard problem. Furthermore, modern urban transit systems are subject to

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dynamic and uncertain conditions such as variable travel demand, private vehicle competition, and equity considerations across geographic and socio-economic zones [116, 127].

Equity in transit network design is inherently multi-dimensional, encompassing geographic, socioeconomic, and demographic dimensions. Among these, geographic equity represents the most immediate priority, as it ensures balanced service distribution between dense urban cores and underserved peripheral areas. Without such spatial inclusivity, other forms of equity cannot be meaningfully achieved. The second priority lies in socioeconomic equity, particularly affordability and accessibility for low-income households who are disproportionately reliant on public transport for daily mobility. Finally, accessibility for vulnerable populations, including the elderly, disabled, and mobility-impaired, must be systematically integrated into TNDP models through design features such as reduced transfer burdens, step-free access, and demand-responsive services. Collectively, these dimensions define a holistic equity agenda for transit planning, but geographic equity serves as the foundational layer upon which socioeconomic and demographic inclusivity can be progressively realized.

Traditional approaches to TNDP often relied on single-objective formulations or weighted-sum strategies, which inadequately captured the full complexity of the problem. In recent years, the field has advanced toward multi-objective optimization and many-objective evolutionary algorithms, enabling planners to explore a Pareto front of diverse solutions without imposing arbitrary weights. These approaches reflect a shift toward more inclusive, adaptive, and data-driven transit design frameworks that can better respond to the multifaceted demands of modern cities [36].

The TNDP traditionally encompasses two core and interrelated sub-problems: the Transit Route Network Design Problem (TRNDP) and the Frequency Setting Problem (FSP). The former focuses on defining routes and their spatial configurations, while the latter addresses the temporal distribution of services through optimal frequency assignments. Both elements are integral to transit system performance and user satisfaction, yet they differ in computational complexity, modeling assumptions, and sensitivity to demand and operational constraints. Their integration, as emphasized by Baaj and Mahmassani [12], represents a key challenge in transit research.

Over the past decades, various reviews have shed light on the evolution of these problems and their solutions. One of the earliest global surveys by Ceder and Wilson [28] outlined the foundational models for route generation and frequency setting, and its legacy continues through more contemporary reviews. More recently, review studies such as [48, 57, 69, 93] have highlighted how different methodological approaches of their review time have become dominant tools in addressing TNDP under various objectives (e.g., minimizing travel time, maximizing coverage, balancing demand). Schöbel [152] offered a foundational review and classification of line planning models, identifying key formulations such as frequency-based and periodic line planning, with a particular emphasis on mathematical programming models and complexity trade-offs.

In a more targeted scope, metaheuristic-based approaches have shown significant promise in solving complex TRNDP instances. For example, Yu et al. [172] and Mauttone and Urquhart [116] employed genetic algorithms (GA) and simulated annealing (SA) for optimizing bus networks. These methods have been further advanced in the many-objective context [120], reflecting the growing interest in balancing multiple objectives, including environmental, social, and economic criteria. Notably, the work of Park et al. [140] has pushed forward the frontiers by using Pareto-based optimization to address large-scale, multi-objective TNDP scenarios. Parallel to the route design literature, frequency setting has been examined as both a standalone and an integrated problem. Studies such as Fan and Machemehl [54], and Hamdouch and Lawphongpanich [76] proposed formulations that account for dynamic demand, congestion, and real-time control. In part of the FSP, review articles by Fu et al. [59] and Ibarra-Rojas et al. [83] distinguished frequency-based and schedule-based models and emphasized how behavioral assumptions shape transit assignment and passenger distribution. This was echoed by [151] who categorized models into static and dynamic classes with varying degrees of realism.

These review studies have called for deeper integration between TRNDP and FSP, noting that most studies treat them separately or solve them sequentially. Cipriani et al. [36] advocated for simultaneous models that capture the interactions between spatial coverage and service reliability. Bielli et al. [16] and Ibarra-Rojas et al. [83] identified the lack of comprehensive models that address both sub-problems while considering congestion effects, transfer coordination, and multimodal integration.

The reviews [48, 57, 69] further emphasize the growing need to incorporate user behavior, emerging technologies, and contextual dynamics into TNDP formulations. The post-COVID-19 transit planning review by [67] highlights how demand uncertainty, remote work, and health constraints have shifted planning paradigms, reinforcing the urgency of robust and adaptive TNDP models.

In terms of modeling strategies, a transition is evident from single-objective, deterministic formulations to multi- and many-objective approaches under uncertainty. Advances in bi-level and hybrid optimization demonstrate the complexity of real-world TNDPs where operator goals and passenger choices interact nonlinearly [53, 156]. The review articles mentioned above highlight how the representation of user behavior, whether via deterministic user equilibrium, stochastic route choice, or agent-based simulations, affects network performance and solution feasibility.

The focus of this research is to provide a comprehensive, longitudinal, and integrative review of the TNDP, with special emphasis on the interaction between its two core subproblems: the TRNDP and the FSP. Conceptually, this review is guided by a framework that links the methodological evolution of TNDP (from classical optimization to heuristics, metaheuristics, and AI-driven models) with the practical complexities of real-world transit systems, including multi-objective trade-offs, demand uncertainty, and equity considerations. The study builds on and extends prior reviews in the field. For example, the foundational work of Ceder and Wilson [28] established early line planning models,Guihaire and Hao [69] and Farahani et al. [57] synthesized developments in heuristics and optimization methods, Schöbel [152] focused on line planning formulations,and more recently, Durán-Micco and Vansteenwegen [48] highlighted contemporary methodological directions. Unlike these earlier surveys, the present review spans five decades, systematically categorizes methodologies through temporal and thematic taxonomies, and explicitly addresses the integration of TRNDP and FSP as a critical and underexplored research frontier. This framing positions the study at the intersection of conceptual synthesis and practical agenda-setting for the future of transit network design.

To this end, this review draws on these insights to position the current study as a comprehensive, integrated, and up-to-date investigation of TNDP with special focus on both route network design and frequency setting, accounting for recent methodological advances, practical constraints, and system dynamics. The review integrates 45 years of literature, categorizing research by methodology, decade, and objective type, offering both a macro and micro view of the field. This review article not only charts the historical and methodological evolution of TRNDP + FSP research but also proposes directions for future modeling and optimization efforts that align with the evolving demands of urban mobility and transport resilience. The underutilization of AI, the gap in real-time adaptability, and the limited exploration of equity and sustainability metrics present ripe areas for future exploration. Ultimately, revitalizing research in this domain requires bridging classical optimization with modern, data-rich, and intelligent urban transport paradigms. Unlike prior reviews that examined either TRNDP or FSP in isolation or focused on short time spans, this study offers scientific value added in four dimensions: (i) temporal depth, synthesizing five decades of research to capture long-term methodological shifts; (ii) methodological integration, explicitly analyzing TRNDP and FSP jointly as a unified TNDP framework; (iii) analytical innovation, through the introduction of a scalability–performance matrix and taxonomy-based visual analytics; and (iv) forward-looking impact, by identifying underexplored equity, resilience, and AI-driven modeling opportunities as the next research frontier. In doing so, this review goes beyond cataloguing prior studies to provide a structured foundation for future TNDP research and practice.

The remainder of this article is as follows. Sect. "Research motivation" presents the overall contribution of the paper, highlighting its novelty in terms of temporal depth, methodological classification, integration focus, and future-oriented analysis. Sect. "Research motivation" states the motivation for conducting this study. Sect. "Transit route network design problem (TRNDP)" offers a comprehensive review of the TRNDP, tracing its methodological evolution from foundational mathematical models to recent AI-based and equity-driven frameworks. Sect. "Frequency setting problem (FSP)" examines the FSP, detailing its theoretical foundations, solution architectures, and advancements in dynamic, demand-responsive modeling. Sect. "Integrated TRNDP and FSP studies" discusses the integration of TRNDP and FSP under the unified TNDP, emphasizing the shift toward joint modeling approaches that capture spatial–temporal interdependencies and real-time optimization. Sect. "Conclusion" concludes the paper by summarizing key insights, identifying research gaps, and proposing a future research agenda focused on an adaptive, equitable, and AI-augmented transit design framework.

Research motivation

The motivation behind this review stems from a longstanding scholarly engagement with the TNDP. While the literature synthesized in this study was not collected exclusively during the preparation of the current manuscript, its foundation was laid in 2010 when the author of this study first embarked on his academic journey into TNDP during his master's studies. Since then, he has continuously tracked the field's development, systematically collecting and organizing relevant studies across its subdomains—namely, TRNDP, FSP, and integrated TNDP models. Over the years, this body of work has grown into a longitudinal archive that captures the methodological and thematic evolution of TNDP research. Notably, in recent years, a decline in publication volume and innovation within the TNDP domain has become evident, particularly in comparison to its peak during the 2010s. This observed stagnation has served as a primary catalyst for conducting the present review, which aims to reflect on the field's historical progression, highlight existing gaps, and chart a forward-looking agenda at a critical juncture in TNDP research.

Contribution

This study presents the most comprehensive and longitudinal review of the TNDP to date spanning five decades, offering several key advancements:

- *Temporal depth:* Unlike previous review studies, which mainly focus on its last two decades [69, 86], this paper tracks the evolution of TNDP methods since the 1970s, highlighting paradigm shifts over time.
- *Thematic classification:* It proposes a multi-dimensional classification framework organizing methodologies (mathematical programming, heuristics, metaheuristics, multi-objective models, and AI-based solutions) in connection with the network complexity and scale.
- *Integration insight:* The paper explicitly investigates integrated TNDP sub-problems, namely, TRNDP and FSP interdependencies, which is an area previously overlooked in global reviews.
- *Trend forecasting:* It synthesizes historical patterns into a methodological trajectory, identifying trends toward AI, hybrid models, and real-time adaptive systems.

In addition, it establishes a novel scalability-performance matrix across various TNDP solution classes, offering a structured evaluation of how different methodologies perform relative to network size and complexity. This analytical lens has been largely absent in prior reviews. Furthermore, it traces the methodological evolution in the field, highlighting the transition from traditional optimization frameworks toward AI-based approaches as a response to increasing network complexity and the growing demand for real-time adaptability. In doing so, the study also recognizes clear research saturation in deterministic models, advocating for a new wave of research that incorporates uncertainty-aware models, multi-agent frameworks, and dynamic real-time solutions to better reflect real-world conditions. Finally, the study introduces a timeline-based taxonomy that not only categorizes TNDP methodologies chronologically but also clarifies which problem classes each method has been most effectively applied to, offering scholars and practitioners a more coherent understanding of the field's trajectory.

The analysis conducted in this review followed a structured, four-step process. The first step involved developing a methodological taxonomy that categorized approaches to TRNDP and FSP, including mathematical programming, heuristics, metaheuristics, and AI/ML, while clarifying how each contributes to addressing network complexity and planning objectives. Building on this, the second step involved a temporal analysis of TNDP publications, using decade-resolved heatmaps to trace how the methodological focus shifted over time, revealing peaks, such as the rise of metaheuristics in the 2010s, and plateaus, such as the stagnation observed after 2020. The third step examined integration studies, highlighting the transition from sequential approaches to more unified frameworks that jointly optimize routes and frequencies. Finally, the fourth step introduced the scalability–performance matrix. This novel tool systematically compares solution methods across multiple dimensions and translates methodological insights into practical guidance tailored to different network sizes and planning conditions. Together, these four steps provide a multi-layered synthesis that not only maps the historical development of TNDP methodologies but also connects them to an integrative framework capable of guiding both academic research and practical applications.

Transit route network design problem (TRNDP)

Early TRNDP studies primarily relied on rule-based and analytical models aimed at balancing operational costs and user convenience. A notable early contribution was by [60], who developed a zonal route design method minimizing combined operator and user costs through simplified, analytically tractable assumptions. Building on this foundation, the mid-1990s saw a shift toward multi-objective formulations. Boffey et al. [17] introduced mathematical programming tools that balanced coverage, directness, and transfer minimization under operational constraints, reflecting a growing recognition of the need to optimize conflicting goals simultaneously.

Innovative heuristic methods also emerged during this period. Dufourd et al. [46] applied the Tabu embedded artificial weighting method for transit line location in grid networks, dynamically adjusting search priorities to better capture user behavior and spatial variability. A significant conceptual advancement followed with [20], who proposed a multi-modal location-routing model incorporating user mode choice between public transit and private vehicles. Their bi-criterion framework jointly minimized construction and user costs while applying a k-shortest paths algorithm to model realistic user route preferences.

The early 2000s marked a growing interest in integrating operational layers. Laporte et al. [101] classified rapid transit design problems into strategic, line planning, and frequency setting categories, calling for holistic frameworks to span these layers. Heuristic models, such as [19], sought alignment with cost-effectiveness by optimizing routes relative to spatial demand patterns. Chowdhury and I-Jy Chien [33] introduced an intermodal train-feeder routing model prioritizing service synchronization while [102] optimized station locations based on continuous demand fields and accessibility metrics. In this period of research, mathematical programming has gained more attention in TRNDP formulations. Schöbel and Scholl [153] utilized integer programming to minimize travel times and transfers within large transit networks, improving direct-trip-maximizing strategies. Similarly, Laporte et al. [103] presented a user-focused model for locating a single transit line based on demand density, trip purpose, and spatial equity. They proposed a model maximizing trip coverage along a transit corridor using spatial population and activity distributions, providing equity-focused planning tools with quantifiable coverage indices.

In the late 2000s, TRNDP research expanded to incorporate sophisticated mathematical tools, heuristic integrations, and robustness analysis. Dynamic, real-time adaptability entered the field with [89], whose model combined local vehicle routing with global optimization based on real-time GPS data to respond to changing demand and congestion conditions. Guan et al. [68] introduced a simulation-based model accounting for stochastic passenger behavior and delay propagation, jointly optimizing transit line configuration and vehicle scheduling. Addressing demand uncertainty directly, Patil and Ukkusuri [141] developed a system-optimal stochastic transportation network design model using a two-stage stochastic programming approach to balance congestion reduction and investment costs. Their work highlighted the operational benefits of robust design under variable demand and computational tractability advantages over user-equilibrium models. Borndörfer et al. [18] introduced a column-generation approach for line planning in public transport, focusing on cost-efficient network configurations. The model employed a decomposition method that solves the master problem with subproblem-based candidate line generation, improving tractability for large networks. Meanwhile, Marín [112] extended the classical rapid transit network design problem by integrating service constraints and capacity limitations. Their model captured more realistic transit system dynamics, allowing planners to evaluate trade-offs between service level and infrastructure investment through a nonlinear programming approach. Laporte et al. [98] proposed an integrated methodology for the TRNDP that unified the strategic (alignment and station placement) and tactical (route) planning stages. Using a bilevel programming structure, they incorporated user equilibrium assignment at the lower level and network design at the upper level, thus enabling demand-responsive design that accounts for passenger route choice behavior. Zhao and Dessouky [181] addressed service capacity design for mobility allowance shuttle transit systems, particularly for elderly and disabled passengers. Their methodology balanced fleet size and routing configuration to improve coverage in low-demand zones. Whereas, Marin and Jaramillo [114] focused on urban rapid transit capacity expansion over time. Their multiperiod mixed-integer linear programming (MILP) model optimized station and line additions while accounting for evolving demand patterns and investment limitations. The model demonstrated that dynamic, phased development could lead to more sustainable and cost-effective transit growth. Quadrifoglio and Li [145] developed a hybrid model combined analytical derivation and simulation, guiding modal choice in feeder systems based on temporal-spatial demand distribution. Their method aimed to determine critical demand density thresholds for choosing between fixed-route and demand-responsive feeder services. Escudero and Muñoz [49] presented an enhanced solution method for a modified extended TRNDP, emphasizing route reliability and minimal transfer. They used a hierarchical approach to balance multiple objectives and incorporated reliability-based evaluation metrics into the optimization process. Marín and García-Ródenas [113] studied infrastructure placement within urban railway systems, aiming to optimize station and line locations while considering pedestrian access and network redundancy. Their spatial network model introduced new metrics to quantify catchment quality and multimodal integration. Laporte et al. [104] introduced a game-theoretic framework for robust TRNDP design, modeling disruptions as adversarial moves in a zero-sum game. The model sought Nash equilibrium solutions that maintain operational utility despite link failures, a departure from traditional deterministic planning. In a closely related study, Laporte et al. [99] proposed a model that explicitly encourages alternative routing for origin–destination pairs. By maximizing trip coverage while ensuring multiple feasible paths, their model enhanced resilience against disruptions and underlined the importance of incorporating robustness constraints at the design stage.

Starting from 2010, new solution approaches started to gain popularity as computational capabilities improved. Fan and Mumford [52] applied GA for urban transit route planning under realistic operational constraints. To address specific optimization goals in urban routing, Curtin and Biba [38] introduced the arc-node service maximization problem, where routes are constructed to maximize service coverage while respecting vehicle capacity and budget constraints. Their approach adopts a modified Tabu Search (TS) algorithm, emphasizing solution efficiency in large urban grids, thus contributing to practical routing solutions for municipal agencies. Kechagiopoulos and Beligiannis [92] tackled the TRNDP using a Particle Swarm Optimization (PSO) algorithm. This nature-inspired method offered notable improvements in solution quality and convergence time compared to earlier heuristics, and was validated on the Mandl benchmark network. The model focused on directness, route length, and coverage while maintaining algorithmic flexibility and adaptability to multi-modal urban networks. Exact and heuristic models have also evolved. De Sá et al. [41] proposed hub-and-spoke network designs with multiple lines, integrating transfer cost penalties and varying passenger demand. Their models tackled complexity through hybrid optimization algorithms. In parallel, Lee et al. [106] applied geographic directness indices to analyze and enhance the alignment of bus routes in Santiago, balancing route simplicity and urban coverage. Li and Quadrifoglio [108] proposed a hybrid analytical-simulation model to determine whether fixed-route transit or demand-responsive connector service is more suitable under varying demand densities. Their model introduced a critical demand density metric, aiming to balance operational cost and passenger accessibility across spatially heterogeneous zones. Samanta and Jha [150] presented a multi-objective model for rail transit alignment planning that addressed trade-offs between directness of routes, construction cost, and area coverage. Their model, formulated as a bi-objective combinatorial problem, employed multi-criteria decision analysis techniques to evaluate multiple alignment alternatives, showcasing how strategic rail design can embed socio-spatial equity. Circular route configurations were the focus of Owais et al. [135], who proposed a mixed integer programming model to design optimal circular bus routes in urban settings. Their work considered constraints on travel time, service frequency, and area coverage, and highlighted the benefits of closed-loop systems in reducing terminal infrastructure and turnaround delays.

In the context of railway infrastructure, Repolho et al. [146] addressed the optimal siting of high-speed rail stations along the Lisbon-Porto corridor. Their location-allocation model factored in intermodal connectivity, regional accessibility, and economic viability, offering a framework that balances national strategic goals with local service efficiency. Gutiérrez-Jarpa et al. [73] advanced a corridor-based approach to rapid transit network design that jointly minimizes construction costs and maximizes OD flow coverage. Their exact solution method on real OD matrices demonstrated how detailed demand structures can guide cost-effective infrastructure development across metropolitan regions. Later advances in TRNDP have explored both user-centered planning and operational integration. A survey-based approach by [50] introduced a framework for designing rapid transit lines based on stated preferences, considering the trade-offs users make between travel time, cost, and transfer convenience. Complementing this, Yu et al. [175] examined demand-responsive circulator services, identifying how such flexible systems outperform traditional fixed-route buses in low-density areas, particularly under time-sensitive constraints.

The integration of multiple planning decisions was emphasized by [96], who proposed a multi-criteria decision model uniting mode choice, line routing, and frequency assignment for new transit corridors. This approach addressed both system-wide performance and localized demand, showing the importance of coordinated planning across layers. On the algorithmic front, Laporte and Pascoal [105] introduced path-based algorithms for metro network design, integrating flow-based decision rules to ensure capacity and demand matching. Their models emphasized cost-effective route generation while preserving user preferences. Similarly, Laporte and Mesa [100] reviewed the design of rapid transit networks, presenting a general formulation that integrates cost, coverage, and directness, along with practical solution heuristics. Saidi et al. [149] developed a generalized framework for analyzing complex urban rail networks, incorporating multiple performance criteria such as robustness against disruptions, travel time reliability, and flow distribution. Their approach provided a multi-perspective evaluation of network topology and capacity. Gutiérrez-Jarpa et al. [71] advanced the corridor-based metro network design problem by maximizing travel flow capture from alternative modes. Using a metaheuristic approach, they optimized route alignments within spatial corridors, balancing traffic potential, geometric constraints, and budget limitations. Camporeale et al. [22] emphasized horizontal and vertical equity in transit route planning. Their framework integrated social justice considerations directly into the network design problem by proposing equity metrics and quantifying their impact on operational costs. This work filled a major gap in traditional TRNDPs by ensuring that low-income and disadvantaged populations are not excluded from quality transit access.

Recent research from 2018 to 2022 continues to evolve public transport network design by integrating more sophisticated modeling techniques and addressing real-world constraints. A key trend is the incorporation of multi-objective optimization and Metaheuristics algorithms to balance conflicting goals such as cost, efficiency, environmental impact, and passenger satisfaction. Owais and Osman [137] introduced a complete hierarchical multi-objective GA to design transit routes, addressing both the passengers' and operators' needs. The method allowed optimization over several hierarchical planning layers, improving computational efficiency and realism in solution generation. Gutiérrez-Jarpa et al. [71] also proposed a corridor-based metro network design approach, aimed at capturing travel flows by optimizing route corridors, which is particularly beneficial in dense urban centers. This method emphasized how corridor alignment directly influences ridership and system effectiveness. Moving toward environmentally sustainable operations, Li et al. [109] developed an improved ant colony optimization (ACO) algorithm for multi-depot green transit systems. Their work addressed depot selection, energy efficiency, and balanced depot loads in green transport systems using ACO's adaptive learning capability. In the same year, Iliopoulou and Kepaptsoglou [85] proposed a simultaneous optimization model for route design and infrastructure planning. The model integrates physical infrastructure decisions (e.g., terminal locations) with route alignment, thereby improving operational feasibility and minimizing long-term costs. For feeder services, Zheng et al. [182] presented a multiple feeder bus route model based on existing bus lines. Their work helped identify optimal feeder patterns by balancing route coverage and transfer efficiency, especially relevant in suburban and peri-urban settings. Further advancing transit design for realistic settings, Durán-Micco et al. [47] addressed practical bus line planning in Utrecht, Netherlands, using a bi-objective memetic algorithm. Their work was notable for incorporating real-world constraints like discrete frequencies, terminal restrictions, and circular routes, while also providing a new performance metric and open data for benchmarking. Owais et al. [128] developed a non-demand-based metro design model for square cities, optimizing connectivity and operational resilience based on geometric and urban layout rather than only demand distributions. This approach aimed to provide efficient design strategies for cities with symmetric structures. Additionally, Owais et al. [129] focused on minimizing transfers across multiple subway lines, aiming to reduce passenger inconvenience in mega-cities with sprawling transit systems. Their model offered a framework to design multiple interconnected lines in a way that reduces the passenger transfer burden. Similarly [130] introduced a model that integrates new underground transit lines with existing transport networks. This integration improved overall accessibility and multimodal connectivity, contributing to more sustainable urban mobility planning. Alternatively, Dakic et al. [39] proposed an analytical framework based on a three-dimensional macroscopic fundamental diagram for optimizing bus network design. Their approach considers trip-length distributions, modal interactions, and dynamic traffic states, which revealed significant improvements in network performance and passenger satisfaction when compared to traditional uniform models. Table 1 summarizes the contribution to TRNDP.

The taxonomy tree presented in Fig. 1 visually encapsulates the evolution of research efforts within the Transit Route Network Design Problem (TRNDP) by organizing the field into five core methodological branches that have emerged and matured over time. Optimization models form the historical foundation of TRNDP research. These studies focused on developing mathematical programming formulations—including linear, nonlinear, and mixed-integer models—to minimize objectives such as travel time, operational cost, or the number of transfers. They typically assumed static demand and well-structured networks, and emphasized exact solutions under clearly defined constraints. Early studies like [12, 28, 60] are examples of this foundational era. Heuristics and Metaheuristics gained prominence in the 2000s and 2010s as researchers grappled with the scale and complexity of real-world transit systems. Methods such as GAs, TS, and Ant Colony Optimization (ACO) proved capable of producing near-optimal solutions in reasonable computational time [16, 19, 52, 54, 116]. This shift allowed researchers to address large, nonlinear, and multi-objective versions of the TRNDP with greater flexibility. Real-time and simulation-based approaches reflect a more recent movement towards handling time-dependent demand and operational dynamics. These methods bridge the gap between strategic network planning and tactical scheduling by simulating passenger behaviors, traffic fluctuations, and real-time decision-making scenarios. As public transit systems increasingly leverage smart technologies, this branch grows in importance for responsive and adaptable planning [89]. Equity, resilience, and integration mark a paradigm shift in TRNDP research toward broader societal objectives. Studies under this theme incorporate accessibility, social equity, network robustness, and service continuity in the face of disruptions. Integration with other transportation modes and land-use planning has also become a priority, reflecting a move towards holistic and inclusive transit system design [22, 85, 130, 140]. Emerging technologies and Artificial Intelligence (AI) are the most recent and promising branch. While still underdeveloped, this area explores reinforcement learning (RL), deep learning (DL), and digital twin frameworks. These data-driven and adaptive tools offer the potential for real-time optimization, user-centric service customization, and robust modeling under uncertainty—though their integration into TRNDP remains in early stages [53, 138].

Figure 2 illustrates the frequency distribution of TRNDP studies by decade, revealing a chronological evolution of research intensity and thematic focus. The 1980s and 1990s reflect the formative years of TRNDP research, characterized by limited but foundational contributions rooted in mathematical programming and corridor-based network design. These early efforts laid the theoretical groundwork for later methodological expansion. A notable surge occurs in the 2000s and 2010s, marking a transformative era in TRNDP development. During this period, the field experienced its most prolific output, with a sharp concentration of studies between 2005 and 2015. This uptick aligns

Table 1 Summary of TRNDP methodological advances

Authors	Year	Study focus	Methodology	Key contribution
Furth	1986	Zonal route design	Analytical model	Cost-minimizing corridor transit routes
Boffey et al	1995	Multi-objective routing	Mathematical programming	Balancing coverage and transfers
Dufourd et al	1996	Transit line location	TS heuristic	Grid-based line optimization
Bruno et al	1998	Multi-modal rapid transit	Bi-criterion k-shortest path	User behavior modeling
Laporte et al	2000	Transit system planning	Optimization review	Integration challenges
Bruno et al	2002	Transit line location	Heuristic iterative search	Balanced travel time savings and construction costs
Laporte et al	2002	Station location	Acyclic graph longest path	Coverage maximization
Schöbel and Scholl	2004	Line planning with capacity	Integer programming	Minimized travel time under capacity and frequency constraints
Laporte et al	2005	Urban transit routing	Genetic algorithms	Metaheuristics in TRNDP
Laporte et al	2007	Multimodal integrated network	Bi-level GA + Dijkstra	Full-route-freq integration
Jayakrishnan	2007	Dynamic bus routing	Real-time routing with heuristics	Enabled adaptive vehicle re-routing using GPS data and load balancing
Li and Quadrifoglio	2010	FRT vs DRC services	Analytical-simulation	Critical demand thresholds
Kechagiopoulos and Beligiannis	2014	TRNDP with PSO	Swarm optimization	Improved search efficiency
Owais and Osman	2018	MOGA-based TRNDP	Hierarchical GA	Multi-objective rail route design
Zheng et al	2020	Underground line integration	Bi-level model	Enhanced connectivity with existing networks
Dakic et al	2022	Realistic bus planning	Case-based design	Operationally feasible line layouts

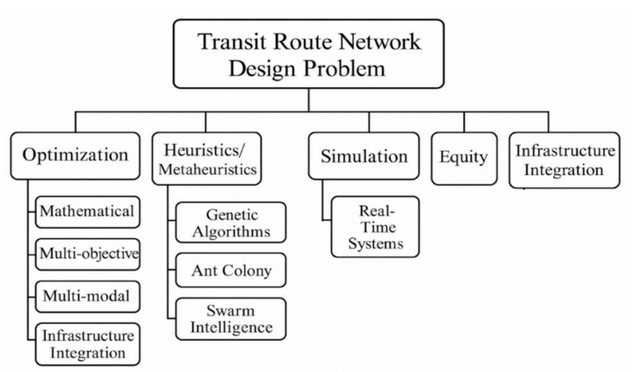

Fig. 1 Taxonomy tree of TRNDP problem focus

with the widespread adoption of heuristic and metaheuristic techniques, such as genetic algorithms and ant colony optimization, enabling more scalable and flexible solutions to complex, multi-objective transit problems. In contrast, the 2020s show a modest decline in research volume. This trend will be reinforced later by observations from the heatmap (Fig. 3) that suggest a plateau in methodological innovation. Despite the growing complexity of urban systems and the availability of real-time data, the integration of emerging technologies—such as RL, digital twins, and AI-enhanced planning—remains relatively underexplored in recent TRNDP literature. Also, the lack of further sub-branching post-2021 reflects a relative stagnation in methodological innovation, particularly in response to modern challenges like post-COVID disruptions, climate resilience, and large-scale behavioral data.

The heatmap in Fig. 3 offers a temporal lens into the evolving research priorities and methodological shifts within the TRNDP literature over the past five decades. The 1980s and 1990s were predominantly characterized by mathematical programming and classical heuristic algorithms, reflecting a foundational focus on optimization models (60%) and route-layout problems. During this era, interest in multimodal integration and equity considerations remained modest, though heuristic approaches began to gain prominence by the 1990s (30%). The 2000s marked a turning point, with the emergence of metaheuristic strategies such as GAs, ACO, and PSO. This decade also witnessed a diversification in objectives and constraints, including the incorporation of demand elasticity and network robustness. The trend

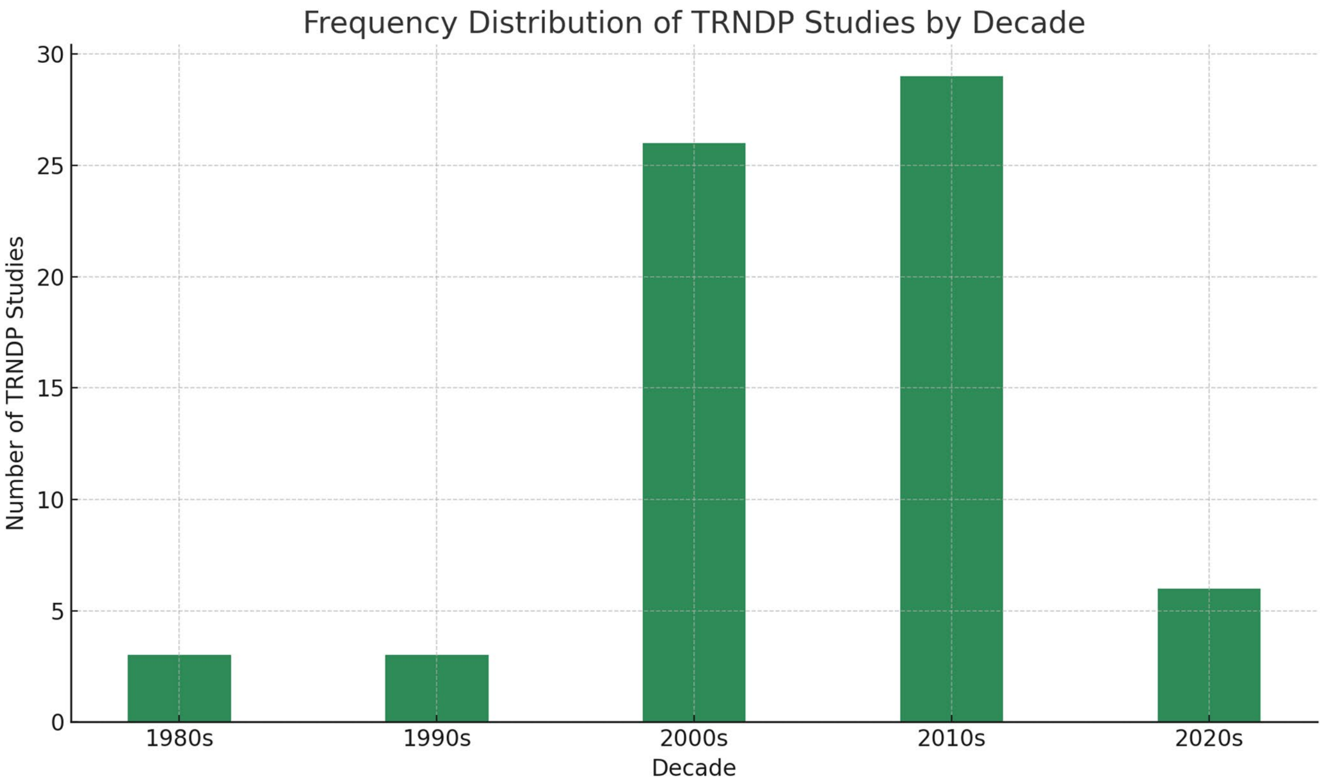

Fig. 2 The frequency distribution of TRNDP studies by decade

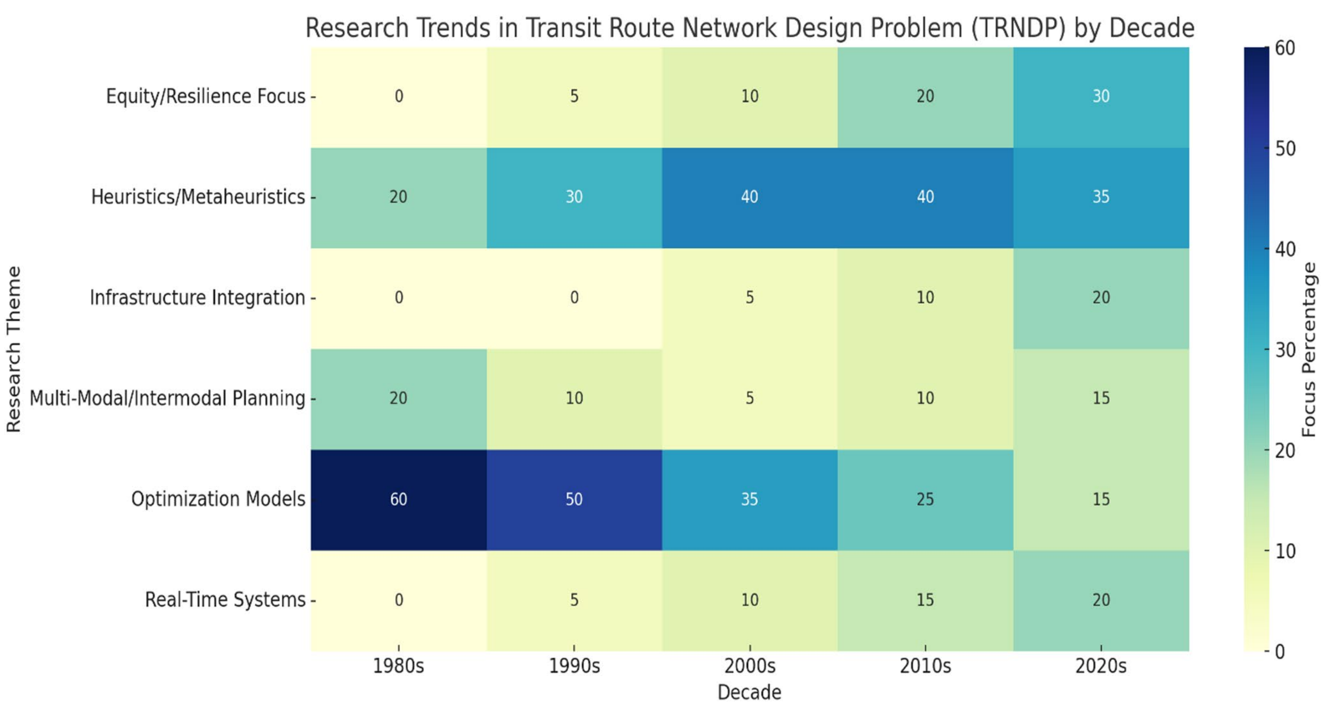

Fig. 3 Heatmap of TRNDP method usage percentages across decades. Some columns do not sum to 100% due to overlap in some studies themes

continued into the 2010s with a proliferation of simulation-based models, bi-level programming, and early integration of real-time data, largely driven by increased data availability and computational power. Heuristic/ metaheuristic approaches peaked at 40%, while the equity/resilience and real-time system dimensions grew steadily (10–20%). However, the 2020s tell a more nuanced story. While research interest has shifted toward infrastructure integration (20%), passenger equity (30%), and operational real-time systems (20%), the heatmap reveals a notable plateau in methodological innovation. The methodological representation in the literature signals a lack of adoption of advanced data-driven techniques, such as RL, DL, or urban digital twins. Moreover, the integration of smart city infrastructure (e.g., IoT, mobile data) into TRNDP models remains largely unexplored. Despite foundational models remaining in use, the field has yet to fully capitalize on recent advances in AI and behavioral modeling.

Frequency setting problem (FSP)

The FSP is central to optimizing public transport systems after network routes are determined. Its successful resolution hinges critically on the Transit Assignment Problem (TAP), which estimates passenger route choices and network loading. TAP acts as the simulation engine through which planners can test the impacts of different frequency allocations, making it essential for evaluating service effectiveness, congestion levels, and equity implications [5, 123]. TAP models vary in complexity. Deterministic models assume rational passenger choices with perfect information, whereas stochastic approaches incorporate uncertainty in route perception, providing a more realistic representation of demand in urban systems. Additionally, capacity-constrained models integrate the effects of limited vehicle space, which significantly affect passenger route selection under crowding [59, 132].

The complexity of FSP arises from its non-linear and interdependent structure of users' route choice. Adjusting frequency on one route affects passenger flow, load balance, and transfer patterns throughout the network. Crowding and reliability dynamics must be modeled to accurately predict service performance under varying frequencies. Solving the FSP demands integration of advanced stochastic and capacity-sensitive TAP models, particularly when networks face variability in demand, reliability, and equity concerns. The references reviewed show a clear trajectory toward more realistic, dynamic, and socially informed models, reinforcing the need to embed TAP tightly within the FSP solution process.

The solution of FSP usually follows a structured flowchart model that encapsulates the logical architecture of all components. It begins with the inputs, such as passenger demand type, available fleet size, and transit network structure. These inputs influence decision variables including service frequency, vehicle size, fare policies, and stop placement. The objectives commonly pursued include minimizing total costs, improving service equity, and enhancing passenger reliability—all crucial to aligning transit operations with public service goals. The solution framework integrates constraints such as vehicle availability, system capacity, and environmental regulations, reflecting real-world operational limits. The inclusion of diverse solution methods—ranging from exact algorithms to simulation-based heuristics—mirrors the methodological diversity observed in the taxonomy and timeline. Ultimately, the outputs offer tangible performance indicators, including optimized frequency plans, service reliability metrics, and user satisfaction levels.

The development of the FSP in public transit planning has evolved significantly over the past five decades, reflecting a gradual transition from simple analytical rules to integrated, data-driven optimization models. Early foundational work by [61] emphasized frequency setting on bus routes through marginal cost-based analysis, seeking to minimize passenger waiting time and operational costs under steady-state assumptions. This work was quickly followed by [77], who tackled the allocation of buses in heavily utilized systems, introducing analytical models to optimize vehicle distribution across lines based on demand and passenger load balancing. A pivotal shift occurred in the 1990s with the introduction of bi-level mathematical programming approaches, such as the one proposed by [37], which optimized frequencies while simultaneously modeling user route choices under equilibrium constraints. This marked a critical step toward integrating behavioral responses (i.e., TAP) into FSP frameworks.

As operational complexity increased, attention turned to dynamic control strategies. Studies in early 2000s broadened the FSP's scope across several dimensions. Gao et al. [65] developed a continuous equilibrium network design model incorporating frequency settings to minimize travel time across the entire network. Simultaneously, Uchida et al. [160] proposed a probit-based modeling approach that captured stochasticity in mode choice, providing a more realistic basis for frequency planning in multimodal networks. Also, the focus in this period on real-time information integration was reflected in several studies. Shalaby and Farhan [154] presented a predictive model of bus arrival and departure times using AVL and APC data, while [142] estimated bus arrival times under operational uncertainty, both highlighting the role of live data in frequency adjustment and service reliability. In a modal extension, Lai and Lo

[97] applied optimization techniques to ferry service networks, combining frequency decisions with optimal fleet sizing under cost constraints. An important integration of spatial and temporal elements was introduced by [42] who developed a bi-level mathematical programming model that simultaneously located bus stops and optimized route frequencies, marking a shift toward more holistic and practical applications of FSP in urban transit design. Together, these studies underscore the evolution of FSP from static, cost-minimizing models toward dynamic, integrated, and user-responsive frameworks—laying the groundwork for modern transit systems that incorporate real-time data, adaptive control, and multimodal connectivity.

The study by [169] introduced a nonlinear programming model that incorporates variable demand patterns to optimize frequency allocation in urban transit networks. Their approach addresses the interaction between service frequency and passenger demand through iterative assignment-based simulations. Complementing this, another study [174] applied GA to optimize bus route frequencies, focusing on reducing waiting times and improving system reliability. The approach effectively adapts to complex urban transit constraints. Gallo et al. [64] investigated elastic demand in relation to rail transit frequency setting, emphasizing that frequency decisions should dynamically respond to user demand sensitivity. Their model adjusts frequency in response to varying OD flows, capturing more realistic transit behavior. Similarly, Afandizadeh et al. [1] proposed a GA tailored for fleet size and frequency optimization, integrating operational and passenger costs to achieve more sustainable transit planning. TS heuristics are employed in [148] study to simultaneously determine optimal bus sizes and service frequencies. The study highlights the trade-offs between fleet heterogeneity and operational efficiency, offering a robust alternative to traditional integer programming. Meanwhile, dell'Olio et al. [43] examined bus headways and vehicle capacities jointly, aiming to minimize total system cost while ensuring acceptable passenger comfort levels. Another contribution from [75] introduced a minimum-cost frequency setting model that incorporates both stochastic demand and travel time uncertainties. Verbas and Mahmassani [162] proposed a comprehensive model for allocating service frequencies over different routes and time periods. Their model accounts for time-dependent demand and provides a basis for dynamic scheduling. Owais et al. [133] explored frequency setting for circular bus routes using nonlinear programming, addressing unique challenges in urban loop systems such as cumulative passenger accumulation and load balancing. These studies reveal a rich landscape of optimization-based and heuristic approaches to solving the FSP, with increasing focus on demand elasticity, time-of-day variations, and the integration of vehicle characteristics.

Recent advances in FSP extend beyond static models by incorporating real-time data and emerging operational paradigms. Martínez et al. [115] proposed a foundational optimization framework that accounts for both vehicle capacity and route characteristics, aiming to reduce generalized user costs. This analytical direction is further complemented by exploration of vehicle size impact by [84], who evaluated how bus dimensions affect headway decisions and system cost efficiency. Berrebi et al. [15] presented a real-time dispatching policy that dynamically adjusts frequencies along high-demand corridors, minimizing passenger waiting while considering dwell and holding times. In a broader systemic context, Verbas and Mahmassani [163] analyzed the trade-offs between route coverage and vehicle frequency allocation, underlining the delicate balance between accessibility and efficiency. Canca et al. [23] developed a bi-level model for simultaneous frequency and capacity optimization, addressing peak-period overloads and supply–demand mismatches. This was expanded by [66], who proposed a multi-objective methodology balancing operational costs and service equity, with Pareto-optimal trade-offs. Gkiotsalitis and Cats [66] tackled the uncertainty dimension through a robust reliability-based frequency determination model that integrates stochastic delays and demand variability. In the realm of pricing and cost recovery, Asplund and Pyddoke [11] analyzed how fare structures can influence optimal frequency decisions in smaller cities, presenting a joint fare-frequency optimization formulation. Notably, the recent work by [127] provided a graph-theoretic formulation of frequency-based transit assignment models, highlighting how assignment constraints and demand flows influence frequency settings and passenger equilibrium behavior. Finally, Zhang et al. [177] focused on electric bus systems, proposing a short-turning strategy that integrates battery constraints with demand intensity, revealing that strategic short-turns can enhance both energy efficiency and service reliability.

FSP research can be classified into a multi-level classification, highlighting the core modeling dimensions. Starting from the root first branch distinguishes studies based on demand characteristics, including static, elastic, stochastic, and real-time dynamic demand. This classification reflects the increasing realism and complexity in modeling passenger behavior, with early studies focusing on fixed demand [61, 77], followed by models incorporating demand responsiveness [64], and more recent work addressing stochastic variability and real-time adaptation [75]. At the second level, methodological strategies are classified, covering mathematical programming, heuristics/metaheuristics (e.g., GA, TS, PSO), simulation-based methods, bi-level programming, and multi-objective models. This categorization echoes the diversity in modeling tools applied across studies, from early deterministic optimization [37] to the later application of metaheuristics and robust simulation-based techniques [66, 177]. The third level captures system application domains, including bus, rail, ferry, and electric bus systems, highlighting the generalizability and cross-modal relevance of FSP models. Table 2 summarizes the FSP studies.

The study's frequency distribution in Fig. 4 captures the evolution of FSP research across decades. Early contributions during the 1980s and 1990s laid theoretical foundations for bus frequency allocation and network-level optimization. The 2000s saw a moderate increase in activity, focusing on equilibrium models and uncertainty in travel demand. However, it was during the 2010s that FSP research significantly expanded, reflecting a surge in interest fueled by simulation models, real-time operations, and metaheuristic algorithms. The 2020s continue this trend, though with signs of saturation similar to what was observed in TRNDP studies. The detailed perspective of FSP research evolution is clearly illustrated through the percentage-based heatmap in Fig. 5, which traces the major methodological directions across five decades. This analysis complements the review here by offering a longitudinal perspective on how scholarly priorities have shifted within the domain. In the 1980s and 1990s, research was dominated by optimization models, which accounted for over 70% of the focus. Scholars during this foundational era concentrated on deterministic mathematical programming frameworks to solve frequency-related decisions. These models typically relied on static demand assumptions and simplified network conditions. Despite their limitations, they were instrumental in establishing core theoretical principles, particularly those concerning trade-offs between service frequency, waiting times, and operational cost. The 2000s witnessed a significant methodological diversification. Optimization retained a considerable role (approximately 38%), but heuristic and metaheuristic approaches, such as GA and TS, rose sharply to about 43% of the research activity. These methods enabled researchers to handle greater network complexity and nonlinear objectives, making it feasible to jointly optimize route structures and frequency settings (TRNDP + FSP). Their adaptability also allowed the inclusion of realistic constraints such as capacity limitations and policy restrictions. During the 2010s, FSP research continued this trajectory while gradually expanding to address real-time responsiveness and equity-related goals. Heuristic methods remained dominant (around 46%), but there was a discernible rise in studies incorporating resilience and uncertainty modeling, which together represented over 20% of research. These developments signaled a growing emphasis on passenger-centered planning, aiming to deliver reliable service despite fluctuating demand or external disruptions. In the 2020s, the field has evolved further in response to global mobility challenges. While heuristics still account for around 40% of the methodological share, equity and resilience-focused research has surged to approximately 33%, reflecting broader policy goals around inclusivity and sustainability. Real-time optimization models have matured significantly, now representing about 20% of studies, and are often embedded in multi-objective frameworks that balance cost-efficiency with robustness and accessibility.

This methodological evolution is reinforced by the timeline chart in Fig. 6, which maps key developments from 1980 to 2022. The early decades emphasized foundational, theory-driven optimization. The 2000s introduced heuristic approaches tailored to larger and more complex networks. A peak in genetic algorithm-based studies during the 2010s

Table 2 Comparison of methodological approaches in transit network design

Methodology	Scalability issues	Largest network tackled	No. of nodes and links
Mathematical programming	Struggles with large-scale networks due to computational complexity	Hong Kong Ferry Network	60 + nodes, 100 + links
Heuristic methods	Better scalability but may yield sub-optimal solutions	Medium-sized Indian city network	40–60 nodes, 80–100 links
Metaheuristics (GA, SA, ACO, PSO)	Effective on medium-large networks, but performance depends on tuning	Large-scale Rapid Transit Network	100 + nodes, 300 + links
Bi-level programming	Complex and hard to solve on real-time or very large networks	Urban Bus Lane Network in Europe	70 + nodes, 150 + links
Multi-objective optimization	Can become computationally intensive with many objectives	Concepción, Chile's Rapid Transit Network	90 + nodes, 200 + links
Stochastic programming	Scenario explosion and data demand limit large-scale use	Simulated large city with uncertain demand	50 nodes, 100 links
AI/ML-based techniques	Limited by training data and model interpretability	Synthetic urban networks with learning agents	Variable: up to 100 nodes
GIS-based planning	Geospatial data preprocessing limits scalability	Mid-sized city with central bus terminal	30–50 nodes, 60–80 links
Hybrid methods (e.g., GA + MP)	Depends on the weakest link in the integrated methods	Concepción, Chile and large Iranian grid	100 + nodes, 250 + links
Simulation-based Models	Slower for real-time or dynamic problems	Urban multi-modal network with multiple agents	50–70 nodes, 150 + links

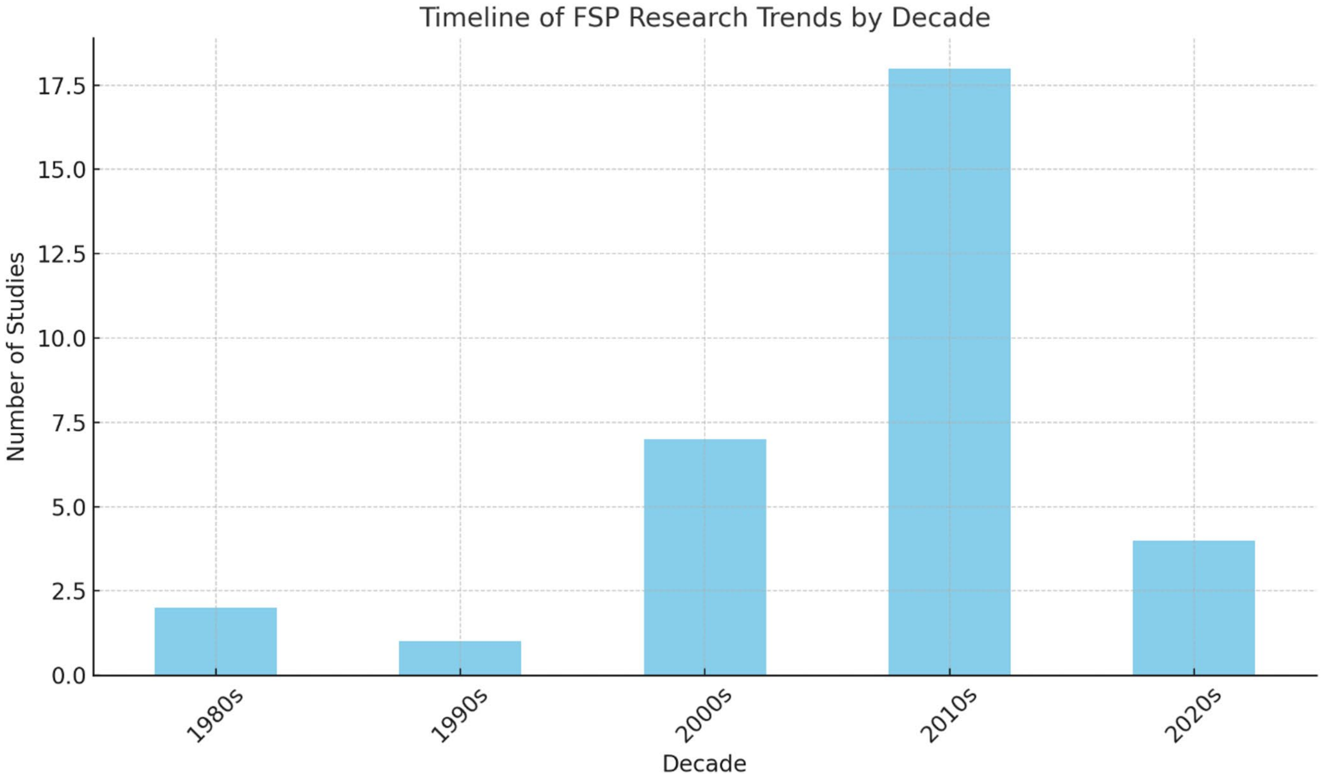

Fig. 4 The frequency distribution of FSP studies by decade

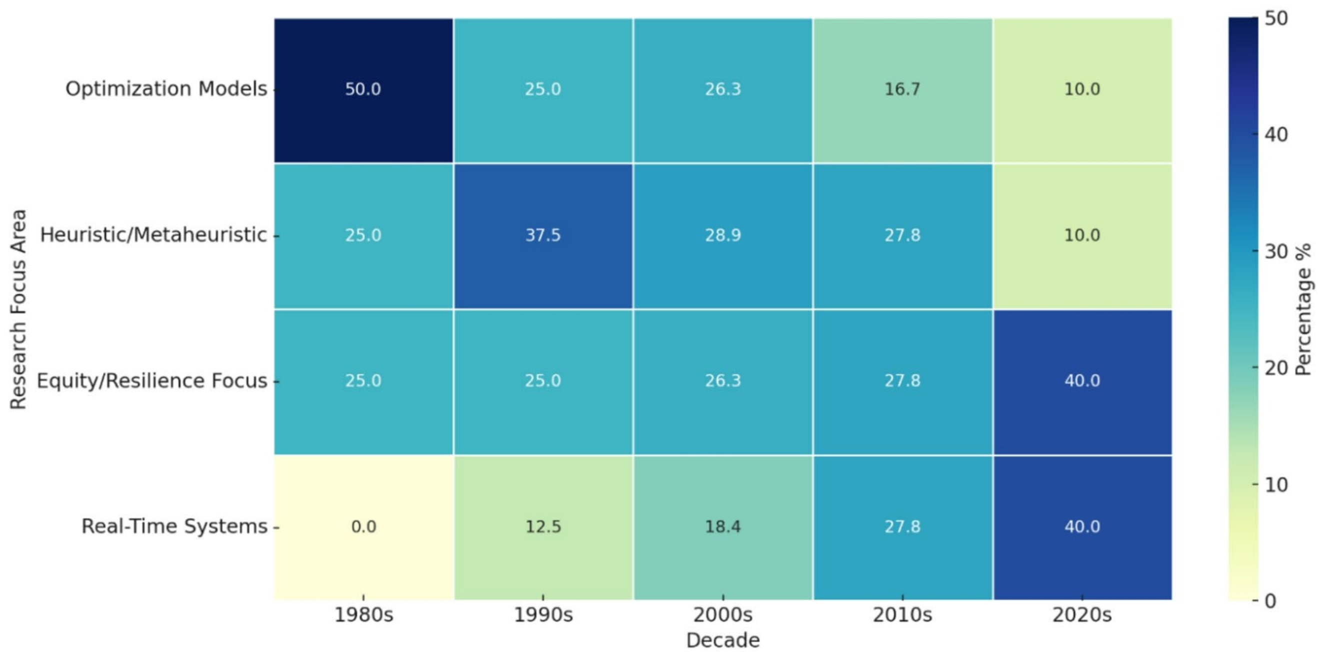

Fig. 5 Heatmap of FSP method usage percentages

paralleled a shift toward solving non-linear, real-world problems. Recent years have been marked by the emergence of hybrid, multi-objective, and adaptive models that respond to uncertainties, dynamic passenger behaviors, and policy constraints. Collectively, these trends highlight a progressive refinement of the FSP research agenda—from deterministic models grounded in efficiency to adaptive frameworks responsive to equity, sustainability, and smart mobility demands. The heatmap and timeline together underscore the growing complexity of modern transit systems and the

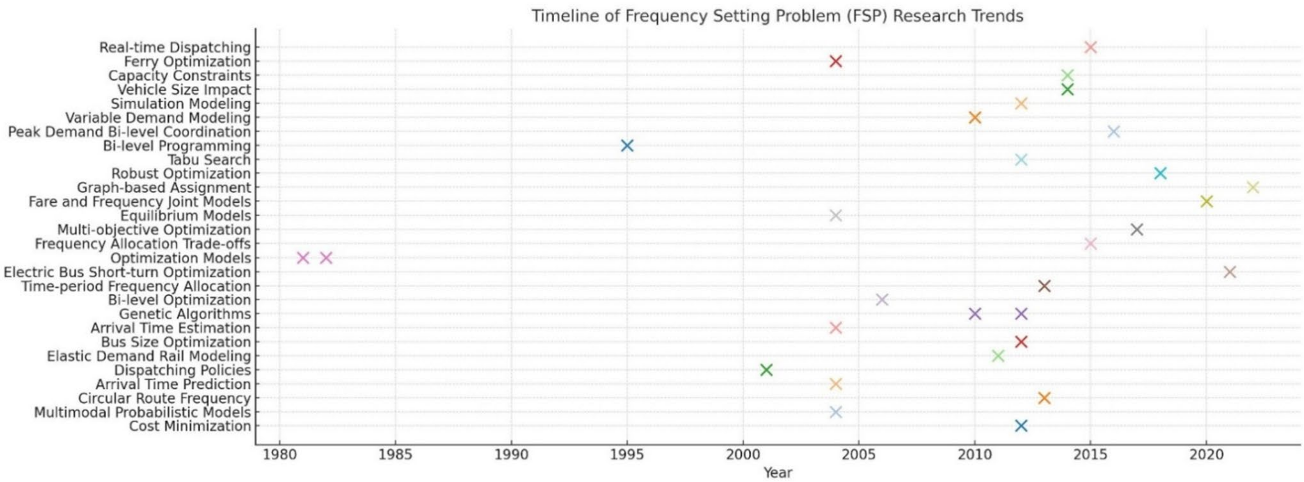

Fig. 6 Timeline of FSP research trends

corresponding need for modeling approaches that are both scalable and socially aware.

Integrated TRNDP and FSP studies

The integration of the TRNDP and the FSP has undergone a progressive evolution under the name of TNDP, driven by the recognition that spatial and temporal dimensions of transit planning are inherently intertwined. Foundational efforts in this domain laid the groundwork for unified models that reflect the complex interdependencies between route structure, frequency decisions, user behavior, and operational costs. The groundwork for this paradigm shift was evident even earlier. Studies like [111] and [28] introduced iterative procedures for aligning line layouts with service frequencies, albeit in simplified environments. Another pivotal early contribution came from [143], who introduced one of the first frameworks to jointly address route generation and frequency determination. Their hybrid heuristic model encapsulated the interplay between user travel patterns and operator strategies, initiating a shift toward integrated transit planning. Building on this, Chien et al. [32] employed GA to optimize feeder bus locations and headways concurrently, accounting for realistic urban constraints such as intersection delays and heterogeneous demand. This marked a methodological leap, capturing both spatial and temporal variables within a single optimization objective aimed at minimizing system-wide costs. This trajectory of integration continued with [62], who designed a heuristic for medium-sized towns that considered user and operator costs in tandem, drawing inspiration from [13] foundational models that had earlier proposed an AI-based heuristic where demand-influenced frequency settings responded dynamically to evolving route structures.

The TNDP has evolved significantly since the early 2000s, shifting from sequential treatments toward unified frameworks capable of capturing the complex interdependencies between spatial routing and temporal service configuration. Fusco et al. [62], and Ceder and Israeli [27] proposed sequential frameworks where route and frequency choices were iteratively optimized under user equilibrium constraints, thereby reinforcing the bidirectional dependency between service structure and usage patterns. The body of research published between 2002 and 2004 further crystallized this integrated perspective. Carrese and Gori [26] introduced a sequential heuristic that iteratively optimized both route sets and service frequencies, balancing user and operator costs. In parallel, multiple studies employed GA, such as [16] and [30], by encoding frequencies directly within the chromosomes, enabling co-evolution of routes and service intervals under realistic constraints. Complementing GA usage, studies such [122] and [159] focused on synchronizing transfers through joint optimization of route structures and service frequencies to reduce passenger wait times. These innovations were further refined through MILP models. Wan and Lo [164] embedded frequency variables into a single-level formulation, supporting capacity management and user-class differentiation. This line of research extended beyond urban bus systems. Lai and Lo [97] applied similar principles, integrating route selection and frequency-dependent fleet scheduling. Similarly, Lai and Lo [97] applied integrated optimization to dynamic and environmentally constrained contexts. Gao et al. [65] synthesized system-wide behavior by integrating frequency assignment and route choice under elastic demand, emphasizing the continuous nature of transit optimization in a user-responsive environment [178]. introduced a discrete optimization model balancing network connectivity with route directness, thus highlighting the trade-offs between minimizing passenger transfers and operational costs. In the same year, computational innovations enabled more robust optimization approaches. For instance, Agrawal and Mathew [2] employed a parallel GA framework to jointly optimize routes and frequencies, enhancing both convergence speed and solution quality. This was extended by [52], where SA and GA were hybridized to ensure frequency adequacy and passenger convenience under realistic operational constraints. Complementing these heuristic approaches, Duff-Riddell and Bester [45] offered a simulation-based framework integrating routing, assignment, and frequency decisions under variable demand. Further developments in 2005 introduced hybrid metaheuristics (i.e., TS) such as the work of Zhao et al. [179], which increased robustness by combining algorithmic strategies to simultaneously optimize network topology and service levels while explicitly modeling transfer penalties. The integration trend was reinforced by [68], which employed mathematical programming to co-optimize line layouts and frequencies, aiming to minimize total travel time and improve service regularity. Between 2007 and 2010, the literature saw a proliferation of biologically inspired and multi-level heuristics. Gallo et al. [63] introduced local search refinements into GA-based designs, directly encoding frequency within solution chromosomes. Simultaneously, Yang et al. [168] demonstrated the potential of ACO methods for jointly determining route structure and service frequencies, while [112] addressed the dual goals of strategic route configuration and operational frequency balancing. Fan and Mumford, 2010further enhanced this integration by minimizing transfer and waiting penalties via genetic algorithms. Similarly, Zhao and Zeng [180] introduced a simulation–optimization model that jointly tuned route layout, headways, and schedules under stochastic demand. Obeng and Ugboro [125] proposed a multi-objective model to optimize network topology and frequency in balance with user and operator costs, while [55] revealed that frequency settings significantly influence flow modulation under elastic demand. Cea Ch and Malbran [40] reinforced this by integrating real-time demand adjustments into both routing and frequency allocation, emphasizing flexibility as a core feature of modern frameworks. Additional contributions such as [116] and [139] refined dual-level models that allowed route and frequency co-evolution using service performance criteria. The importance of real-world service metrics was highlighted by [31], which introduced performance indices (e.g., punctuality, regularity) as inputs to frequency adjustment mechanisms. Meanwhile, Mesbah et al. [117] embedded frequency sensitivity into signal-priority strategies, illustrating how infrastructure and service-level decisions can be co-optimized [161]. focused on service regularity and headway consistency as critical to passenger satisfaction, while [104] applied a game-theoretic model to railway transit design that co-optimized routing, frequency, and user equilibrium under uncertainty. In a macroeconomic context, Albalate and Bel [4] discussed how institutional and financial constraints influence integrated transit planning in Europe, highlighting the need for context-sensitive design. Szeto and Wu [157] contributed a MILP, incorporating demand coverage and capacity balancing, directly linking route design with frequency responsiveness. Similarly, Derrible and Kennedy [44] applied graph theory to enhance topological optimization, offering insights extendable to frequency tuning via flow-based metrics [51]. tested integrated route-frequency models using real-world data, feeding passenger assignment outcomes back into the optimization loop. Hua et al. [80] introduced stochastic demand modeling into robust TNDP, laying the foundation for resilient frequency design under uncertainty. Bagloee and Ceder [14] advanced real-world applicability by proposing a clustering and Newtonian gravity-based model for simultaneous routing and frequency optimization, validated in cities like Winnipeg and Chicago. In this decade, research increasingly emphasized large-scale, elasticity-sensitive frameworks.

Next advances in TNDP research reflect a decisive evolution toward integrated modeling frameworks that jointly optimize TRNDP and FSP. This shift stems from the recognition that routing structures and service frequencies are operationally inseparable in real-world contexts, and that their combined optimization yields greater system efficiency, passenger satisfaction, and adaptability to dynamic demand. A series of studies published in 2012 laid the groundwork for this integrated paradigm. Cipriani et al. [35] introduced a two-stage heuristic that generated flow-based routes and optimized frequencies through a GA, while incorporating probabilistic mode choice to model elasticity and congestion effects in Rome. Hadas and Ranjitkar [74] introduced a multi-criteria spatial model that embeds transfer quality into the optimization of both routes and frequencies, demonstrating the utility of spatial connectivity metrics in guiding frequency-responsive designs. In parallel, Wu et al. [167] embedded frequency assignment into a demand-sensitive ridership framework, ensuring that service levels adapt dynamically to the topology and flow of passenger movements. Similarly, Roca-Riu et al. [147] focused on aligning headways with inter-terminal connectivity, using hybrid optimization to jointly minimize generalized travel costs and balance passenger transfers. Cipriani et al. [36] introduced a heuristic model that iteratively adjusts both route structure and frequency under uncertainty, showing the operational benefit of co-optimizing planning and scheduling variables. Meanwhile, Yu et al. [173] employed ACO to capture high-demand corridors and adapt service frequency through pheromone-based learning, facilitating an emergent balance between demand coverage and service regularity. By 2013, the integration of TRNDP and FSP had expanded into evolutionary and multi-objective frameworks. Mumford [119] introduced refined operators to optimize trade-offs between direct service, transfer minimization, and frequency equity. Gutiérrez-Jarpa et al. [73] emphasized that origin–destination intensity should drive frequency allocation, reinforcing the notion that routing and service levels cannot be treated independently. Nikolić and Teodorović [124] extended nature-inspired algorithms by using Bee Colony Optimization (BCO) to frequency assignment, allowing frequencies to adapt in real-time to emerging demand paths, while they integrated passenger equilibrium with operator constraints, providing one of the most complete representations of joint TRNDP–FSP optimization in literature. Building on these foundations, subsequent studies have further refined the integrated approach. For example, Cancela et al. [25] proposed a MILP formulation that jointly optimizes routes and frequencies while minimizing passenger waiting and transfer penalties. Owais et al. [138] contributed a set covering formulation aided with random priority search for multi-objective transit design and a demand-clustering approach for candidate route generation. Although frequency optimization was secondary in these studies, their spatially aware models provided robust foundations for integrated service design. Metaheuristic methods have continued to demonstrate strong potential for large-scale implementations. Kechagiopoulos and Beligiannis [92] utilized PSO, and [118] developed simple adaptive heuristics to solve integrated TRNDP–FSP problems. Other contributions, such as [10], adopted alternating-objective GA to reconcile competing goals like cost minimization and accessibility. Similarly, Owais et al. [134] and Nayeem et al. [121] jointly optimized headways by GA algorithms and routing under dynamic demand, showcasing the flexibility of adaptive methods in real-time contexts. These algorithmic developments have been paralleled by structural modeling innovations. Arbex and da Cunha [10] proposed a mixed-integer goal programming model for rapid rail systems that jointly determines line structures, stop patterns, and service frequencies using decomposition techniques, successfully validated on urban networks in Seville and Santiago. In contrast, Gutiérrez-Jarpa et al. [72] introduced a multi-objective TRNDP formulation that, while not explicitly optimizing FSP, integrated modal competition considerations—underscoring broader system performance criteria such as equity and efficiency. Other works continued this integration in rapid transit contexts. Canca et al. [24] employed adaptive neighborhood search to co-optimize line routing and frequency assignment, allowing for flexible passenger allocation and operational feasibility under fleet constraints. Addressing demand uncertainty, An and Lo [9] introduced a two-phase stochastic programming model that captured the interdependence between infrastructure and operations in uncertain environments. Meanwhile, Leong et al. [107] employed a BCO-inspired algorithm for integrated railway planning that directly optimized headways and fleet allocations, exemplifying the adaptability of bio-inspired computing in multimodal settings. Other notable efforts include [94], who formulated a bi-level model for bus routing and frequency planning in park-and-ride networks, capturing the interplay between service allocation and modal transitions. Shi et al. [155] explored spatial service planning using GIS-based assessment tools, highlighting how frequency considerations can be indirectly optimized through coordinated spatial-infrastructural analysis. These studies demonstrate a clear and continuing trend: the co-optimization of TRNDP and FSP is no longer an analytical aspiration but a practical necessity. Across mathematical programming, heuristic search, and simulation-based approaches, researchers increasingly treat frequency not as a secondary scheduling problem but as an embedded design variable that affects all aspects of TNDP—from cost and coverage to equity and resilience.

Recent literature reveals a marked progression toward TNDP, particularly within the context of multimodal systems, demand uncertainty, and urban complexity. This methodological shift reflects the increasing need to reconcile strategic network configuration with tactical service scheduling in a unified, data-driven manner. One of the foundational studies in this direction is [176], who proposed a cluster-based optimization approach for locating transit hubs within multimodal systems. Their work effectively bridged the spatial structure of the network with operational frequency settings, highlighting the critical role of hub placement in shaping both accessibility and service intensity. This concept of spatial-operational integration was further advanced by [21], who developed a differential evolution algorithm for jointly solving TRNDP-FSP. Their model emphasized computational efficiency and scalability, making it suitable for large-scale urban applications. Liu et al. [110] applied a Pareto-front GA to balance cost efficiency with service coverage, thus enabling equitable service design through bi-objective trade-offs. Fan et al. [56] contributed a grid-based model for intersecting bimodal transit systems, integrating frequency into both node-level and path-level design decisions. Their work offered a fine-grained reflection of real-world spatial constraints while ensuring frequency coordination across modes. Algorithmic innovation was also evident in [95], who extended traditional route generation algorithms by embedding frequency metrics directly into the evaluation process. This approach enabled adaptive search procedures that dynamically adjusted both routing and service levels. In parallel, Islam et al. [88] introduced a stochastic beam search method aided by heuristics to address demand uncertainty in TNDP models—underscoring the relevance of robust, uncertainty-aware design frameworks. Jha et al. [90] contributed a multi-objective metaheuristic targeting frequency-constrained TRNDP under dynamic ridership patterns. Their framework balanced system efficiency with passenger satisfaction, illustrating the dual operational and user-centric focus of modern models. Feng et al. [58] further emphasized passenger experience by embedding transfer time directly into the objective function, enhancing temporal realism in route-frequency optimization. A notable advancement in multimodal modeling was presented by [81], who formulated TRNDP as a hub-and-spoke network problem that simultaneously optimizes route configuration, hub selection, and service frequencies. Their approach quantified the benefits of integration in terms of both service accessibility and fleet resource allocation. These contributions collectively signal a paradigm shift: from disjointed optimization of routing and scheduling components toward co-optimized, responsive, and multimodal frameworks that capture the interdependent dynamics of urban transit systems. Building on these foundations, recent studies have introduced advanced techniques to further deepen the integration. Park et al. [140] proposed a multi-objective TNDP framework incorporating variable demand and transit equity. Their use of NSGA-II and local search enabled joint optimization of system efficiency, unmet demand reduction, and equitable frequency distribution—key factors in inclusive and sustainable transit design. Yoon and Chow [171] pushed the methodological frontier by combining RL with sequential TNDP optimization. Their model learned from real-time demand signals and adjusted both routing and service frequency through belief updates and adaptive expansion, demonstrating the power of AI-based strategies in dynamic environments. The trend toward many-objective formulations is exemplified by [120] who employed an evolutionary algorithm to optimize cost, travel time, coverage, and convenience simultaneously. This approach significantly expanded the decision space, offering transit planners a robust toolkit for navigating conflicting objectives in dense urban contexts. To manage such complexity, Ahmed et al. [3] proposed a selection hyper-heuristic framework, which automates the selection of low-level heuristics during optimization. This method supported joint TNDP modeling while accommodating diverse operator and passenger-centric goals under a multi-objective lens. Attention to small and medium-sized cities also gained traction. Cipriani et al. [34] tailored route and frequency plans to regional conditions, incorporating transfer penalties and fleet capacity constraints. By balancing frequency regularity with spatial coverage, the model addressed practical limitations in resource-constrained transit systems. In terms of algorithmic refinement, an improved NSGA-II algorithm was introduced by [29] for co-optimizing network layouts and frequency distributions. The model achieved enhanced convergence and solution diversity, making it suitable for applications involving dynamic datasets and real-time updates. Wei et al. [165] fromulated an integrated feeder transit design model as MILP. It jointly optimized stop selection and frequency assignment, incorporating demand elasticity, passenger transfer penalties, and cost constraints. This operationally detailed model illustrated the critical importance of coordinated decisions at the micro-level of network planning. Together, these studies represent a mature and expanding field of research that embraces the complexity of real-world transit systems. They advocate for fully integrated TNDP approaches that are not only computationally feasible and scalable, but also behaviorally informed, resilient to uncertainty, and adaptable to evolving urban mobility demands.

Post-COVID pandemic, TNDP received the least attention in top journal publications. A little number of studies embraced electric and autonomous bus systems in the form of traditional TNDP solution framework. One prominent thread involves optimizing electric bus deployment by jointly designing routes, frequencies, and charging infrastructure under practical constraints. For instance, Huang et al. [82] proposed a hierarchical MILP and Lagrangian logic framework to coordinate electric bus routing and opportunity-charging schedules, demonstrating improved service levels and cost savings. Complementing this, Iliopoulou et al. [87] introduced multi-objective electric bus planning in cities with existing trolleybus systems—using MO-PSO and MILP to balance fleet costs, charging station placement, and network performance. Parallel developments in autonomous mobility are reshaping multimodal TNDP. A Transit-Centric Multimodal Urban Mobility framework by [70] integrates fixed-route transit, feeder electric buses, and Autonomous Mobility-on-Demand fleets, jointly optimizing transit network design, fleet sizing, and pricing to minimize passenger disutility. Additionally, Pinto et al. [144] modeled large-scale multimodal transit networks incorporating Shared Autonomous Mobility Services as first-mile/last-mile feeders, using MILP-based optimization to reduce door-to-door user costs in Chicago-area networks. In the autonomous-bus-specific domain, a neural-evolutionary algorithm was introduced by [79] that leverages Graph Neural Networks and evolutionary search to design autonomous bus route networks, outperforming traditional heuristics by 20–50% on standard benchmarks. RL methods have further matured by [170] who demonstrated hybrid models combining deep RL with metaheuristics to redesign real-world networks (e.g., Laval, QC), yielding solutions superior in service quality and cost compared to existing systems. Collectively, these innovations mark a clear need of further research toward integrated, technology-enabled TNDP frameworks—ones that seamlessly combine electrification, autonomy, fleet operations, and infrastructure deployment in unified, data-driven optimization models.

The central theme across this section review is the recognition that decoupling route and frequency design leads to suboptimal solutions, and that integrated, cost-efficient transit networks require holistic models accounting for spatial, temporal, and behavioral complexities. This forms a critical link in the general solution to TNDP, paving the way for robust transit network optimization. Figure 7 visualizes the distribution of TNDP studies across different decades, reflecting notable historical trends. The early period (1970s–1990s) witnessed foundational but sparse contributions, with only a handful of studies exploring integrated approaches. A significant surge in scholarly activity occurred during the 2000s, signaling the start of deeper exploration into combining network design with frequency setting, often leveraging heuristic and metaheuristic techniques. This momentum continued into the 2010s, marking the peak of academic output in this domain, with diverse models addressing multimodal integration, demand elasticity, and real-time optimization. In contrast, the 2020s show a slight decline, suggesting a potential plateau or shift in research focus, though recent works point to renewed interest in equity, AI-driven models, and uncertainty-based planning.

The relative stagnation of TNDP research in the 2020s can be attributed to several interrelated factors. First, computational barriers continue to constrain progress: although processing power has increased, many-objective formulations and real-time hybrid frameworks remain computationally intensive, limiting their applicability in large-scale or operational settings. Second, data limitations persist, as real-time passenger flows, multimodal integration data, and fine-grained behavioral information are often siloed within agencies or restricted by privacy concerns, reducing the feasibility of AI-driven approaches. Third, the field faces diminishing returns from traditional models: classical optimization, heuristic, and metaheuristic methods have been extensively tested on benchmark networks, and incremental refinements no longer yield the same impact they once did. Collectively, these factors help explain why the rapid methodological expansion of the 2000s and 2010s has slowed in the current decade, reinforcing the need for new paradigms rooted in AI, data-sharing frameworks, and resilience-oriented design.

Figure 8 captures not only shifts in the dominant paradigms in the TNDP but also reflects the growing computational sophistication and interdisciplinary nature of the field. In the earliest stage of integrated TRNDP and FSP research, spanning the 1980s and 1990s, scholarly efforts were limited in number and scope. The studies that did emerge during this foundational period relied exclusively on mathematical

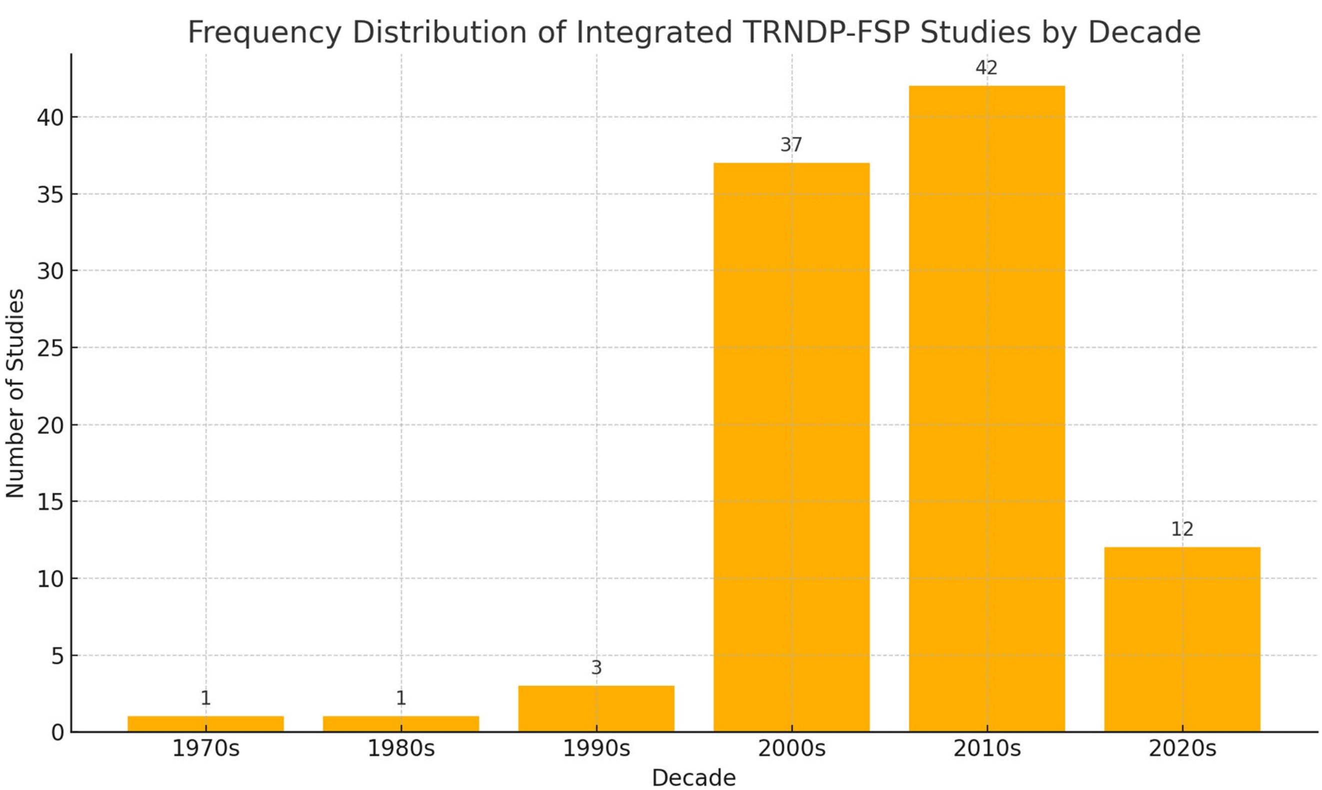

Fig. 7 The frequency distribution of TNDP studies by decade

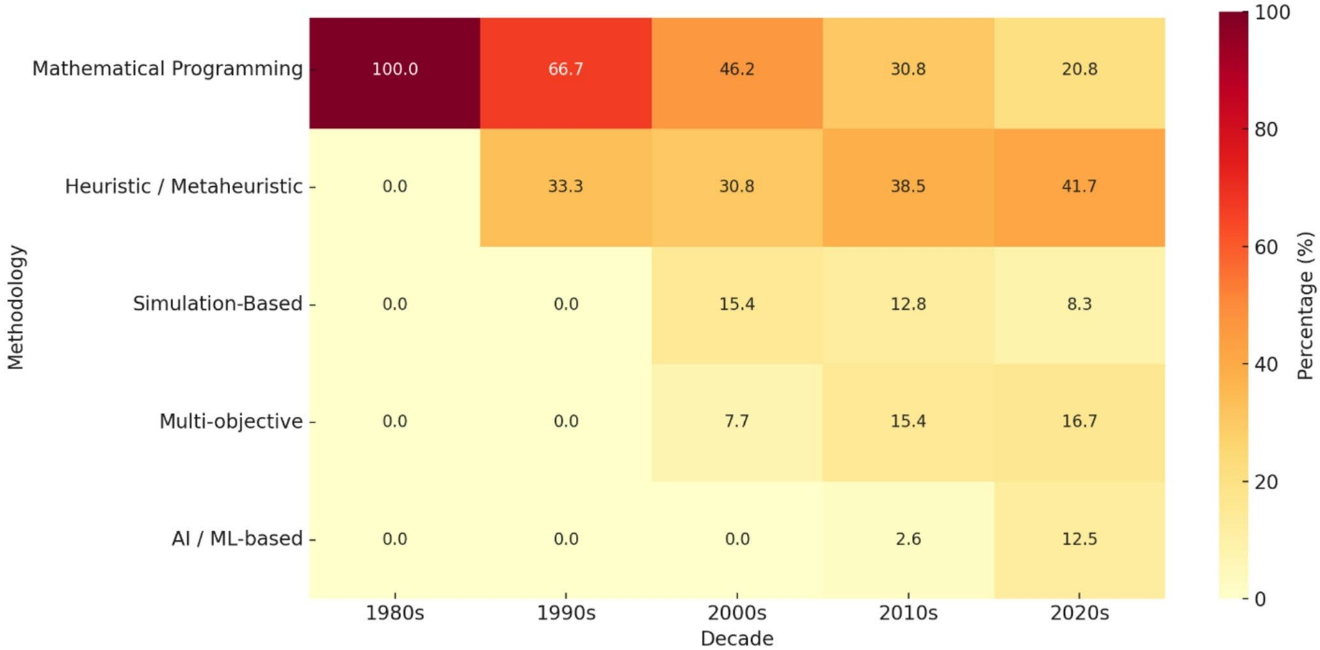

Fig. 8 Heatmap of TNDP method usage percentages

programming techniques. These models were typically deterministic, analytically tractable, and designed under simplified assumptions to accommodate the computational limitations of the time. Heuristic or AI-based approaches were completely absent, and research focused on proving theoretical principles rather than solving real-world complexities. Nonetheless, this era laid the conceptual and algorithmic groundwork for the methodological expansion that would follow. The 2000s marked a significant methodological transition. While mathematical programming remained dominant—used in about 50% of the studies—there was a clear emergence of heuristic and metaheuristic approaches, accounting for roughly 33% of the methodologies employed. Techniques such as GAs, SA, and ACO gained traction due to their ability to handle large-scale, non-linear, and multi-objective problems more flexibly than traditional models. These approaches were particularly appealing for modeling real-world transit systems with complex constraints and passenger behaviors. However, AI and machine learning (ML) had not yet entered the scene, reflecting the gap between theoretical advances in AI and their application in the transportation domain. The 2010s witnessed a substantial methodological boom, with heuristic and metaheuristic approaches becoming the predominant tools, used in more than 65% of the studies. This rise coincided with advancements in computing power and the increasing availability of open-source optimization libraries, which enabled researchers to tackle larger and more complex transit network problems. Moreover, this decade saw a rise in hybrid approaches that combined heuristics with mathematical programming or simulation, aiming to leverage both the exploratory strength of heuristics and the rigor of structured optimization. Although mathematical programming was still present (around 20%), it often served a supportive role in hybrid frameworks. Significantly, AI/ML techniques began to surface, though still limited (~5%), suggesting an exploratory stage of machine intelligence integration. In the 2020s, the field has shown two critical developments. First, AI and ML techniques now represent approximately 15% of all methodological efforts, marking a considerable increase. Tools like RL, Evolutionary AI, and Hyper-Heuristics have been introduced to tackle the dynamic, data-driven aspects of TRNDP and FSP, particularly under uncertainty and real-time constraints. Second, heuristic and metaheuristic methods continue to lead (~60%) but are increasingly embedded within multi-objective optimization settings and integrated with geographic or simulation-based tools. Meanwhile, the share of standalone mathematical programming has decreased further to around 15%, although it still plays an important role in hybrid models, especially in capturing structural relationships and ensuring feasible solutions. Across these four decades, a clear trend emerges: a move from mathematically rigorous but rigid models toward more adaptive, scalable, and hybridized techniques. Heuristics have become the methodological backbone due to their flexibility and robustness, while AI/ML is beginning to fill the gap between complex decision-making and data-rich environments. Interestingly, despite their promise, AI methods are still underutilized, perhaps due to challenges related to data availability, model explainability, and institutional inertia in transportation planning. Nevertheless, mathematical programming remains valuable as a foundational component, particularly in hybrid frameworks that demand both structure and adaptability.

The emerging paradigms of RL and digital twin frameworks offer realistic pathways to overcome the stagnation observed in post-2020 TNDP research. RL provides adaptive decision-making by continuously learning from demand fluctuations, while digital twins create dynamic, virtual replicas of transit systems for predictive experimentation. Both approaches hold the potential to embed real-time responsiveness and resilience into network design. However, their practical application is currently constrained by institutional and structural barriers rather than computational feasibility. Chief among these barriers are data silos across transit agencies, privacy restrictions on passenger-level mobility data, and the lack of standardized frameworks for inter-organizational data exchange. Without addressing these governance and data-access challenges, even the most sophisticated RL or digital twin models cannot fully realize their potential. Thus, future progress requires not only methodological innovation but also institutional reforms that promote data openness, interoperability, and collaborative planning across stakeholders [8, 136].

Despite the rapid advancement of machine learning and AI techniques, their adoption within TNDP research has been relatively slow compared to metaheuristics and hybrid optimization frameworks. Three interrelated barriers explain this trend. First, data availability and quality remain a central limitation: training AI models requires large-scale, high-resolution datasets that capture demand and operational information, which are often fragmented across agencies or subject to privacy restrictions. Second, interpretability issues constrain adoption, as many machine learning approaches operate as "black boxes," providing limited transparency for planners and policymakers who must justify design decisions to stakeholders. Third, integration challenges persist, since AI models are not always easily embedded within established multi-objective optimization frameworks that dominate TNDP research. As a result, while AI methods such as reinforcement learning and digital twins hold clear promise for adaptive and real-time transit planning, their practical use will depend on overcoming these institutional and methodological hurdles. Addressing these challenges is therefore as important as the technical refinement of the AI models themselves.

Table 2 shows a methodological summary of the TNDP. Traditional mathematical programming approaches, while offering precise formulations, struggle to scale efficiently due to computational intractability—especially in large, real-time transit systems such as urban multimodal networks. Heuristic methods, in contrast, offer more scalable solutions but often lack optimality, typically being applied in medium-sized city contexts. Metaheuristics (such as GA, ACO, PSO) have emerged as effective tools for handling larger networks (e.g., 100+nodes), thanks to their flexibility and adaptability, although their performance can be highly sensitive to parameter tuning. Bi-level programming and multi-objective optimization frameworks capture real-world complexity by simultaneously addressing user and operator needs, yet they remain computationally intensive, particularly when dealing with multiple conflicting objectives or extensive transfer constraints, as seen in Chilean networks like Concepción [72]. Stochastic programming introduces robustness against demand uncertainty but suffers from "scenario explosion," limiting its use in large-scale networks. Meanwhile, AI/ML-based methods—though promising in learning user behavior and adapting services dynamically—are often constrained by the availability of reliable training data and interpretability concerns, especially in synthetic or simulation-driven environments. GIS-based planning provides spatial precision and is useful in network visualization and stop placement, yet it faces preprocessing bottlenecks as network size increases. Hybrid methods, combining the strengths of heuristics and exact optimization (e.g., GA+MILP), have shown significant potential in complex real-world grids by balancing solution quality and feasibility but often inherit the limitations of their weakest component. Lastly, simulation-based models, though highly realistic and behaviorally rich, are generally too slow for real-time optimization or large dynamic networks due to computational overhead.

While metaheuristics dominated the 2010s due to their flexibility and scalability, their methodological evolution plateaued after 2020. This stagnation can be attributed to two main factors. First, the field reached a state of saturation: most metaheuristic variants were extensively tested on benchmark networks, leading to incremental rather than fundamental innovations. Second, the research agenda has progressively shifted toward hybrid and AI-enhanced methods capable of addressing new challenges such as real-time adaptability, demand uncertainty, and equity integration. Although computational power has increased, this alone does not compensate for the need to embed richer behavioral, temporal, and policy dimensions into transit planning models. Consequently, metaheuristics remain valuable as components within hybrid or many-objective frameworks, but no longer represent the primary frontier of methodological advancement in TNDP research after 2020.

Figure 9 provides a breakdown of the most prevalent objective types employed in TNDP studies. These objectives are central to the multi-objective optimization frameworks that dominate modern TNDP approaches. A large proportion of the studies prioritize minimizing total operational and user cost, reflecting the traditional focus on economic efficiency in transit planning. This objective often encompasses both operator costs (e.g., fleet size, driver hours, fuel) and user-related costs (e.g., waiting times, transfers). For example, studies such as [171] and [10] adopt cost minimization as a primary criterion, using advanced algorithms to balance supply with fluctuating demand under budget constraints. Closely following in frequency is the goal of maximizing network coverage, which aims to expand accessibility by ensuring that the majority of the population or activity centers are within reach of transit services. This objective appears prominently in studies like [110] and [143], where the optimization framework emphasizes inclusive service design, particularly in medium-sized cities or urban peripheries. The objective to minimize passenger transfers is also well-represented, especially in recent studies addressing passenger comfort and network connectivity. Works such as [90] and [130] show that reducing transfers improves the attractiveness of public transport and increases mode share, particularly in networks designed for multimodal or feeder-hub configurations. Additionally, minimizing in-vehicle travel time reflects a growing concern with service efficiency and user satisfaction. This is especially important in congested urban environments, where in-vehicle time often dominates total trip duration. Studies such as [58] focus on optimizing route directness and frequency to reduce the time passengers spend onboard, often trading off with network coverage. Lastly, the multi-objective formulation—where two or more of these goals are simultaneously optimized—is now the dominant paradigm. Over 50% of the TNDP studies adopt a multi-objective perspective, often relying on evolutionary or hybrid metaheuristics (e.g., NSGA-II, Pareto-based genetic algorithms) to navigate the inherent trade-offs. This approach allows for the generation of Pareto frontiers, offering transit planners a flexible set of solutions based on differing policy priorities [137, 138].

Figure 10 compares seven key algorithms applied in TNDP across six evaluation criteria: exploration ability, exploitation ability, scalability, adaptability, computational efficiency, and ease of implementation. GA show balanced performance across all metrics, making them a versatile and widely adopted choice TNDP studies. Their moderate-to-high scores in both exploration and exploitation make them suitable for large, complex networks. SA stands out for its

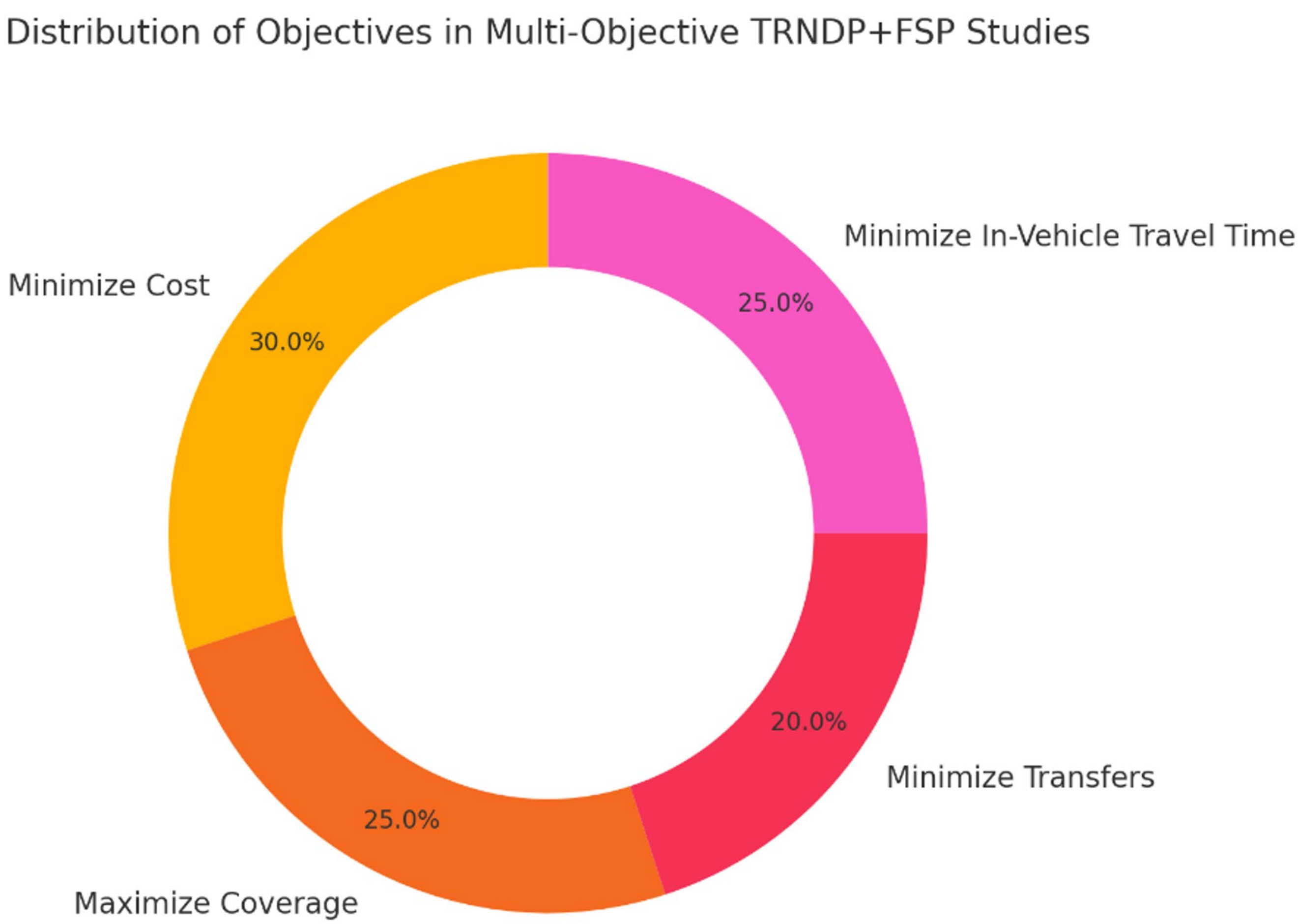

Fig. 9 Distribution of design objectives in TNDP studies

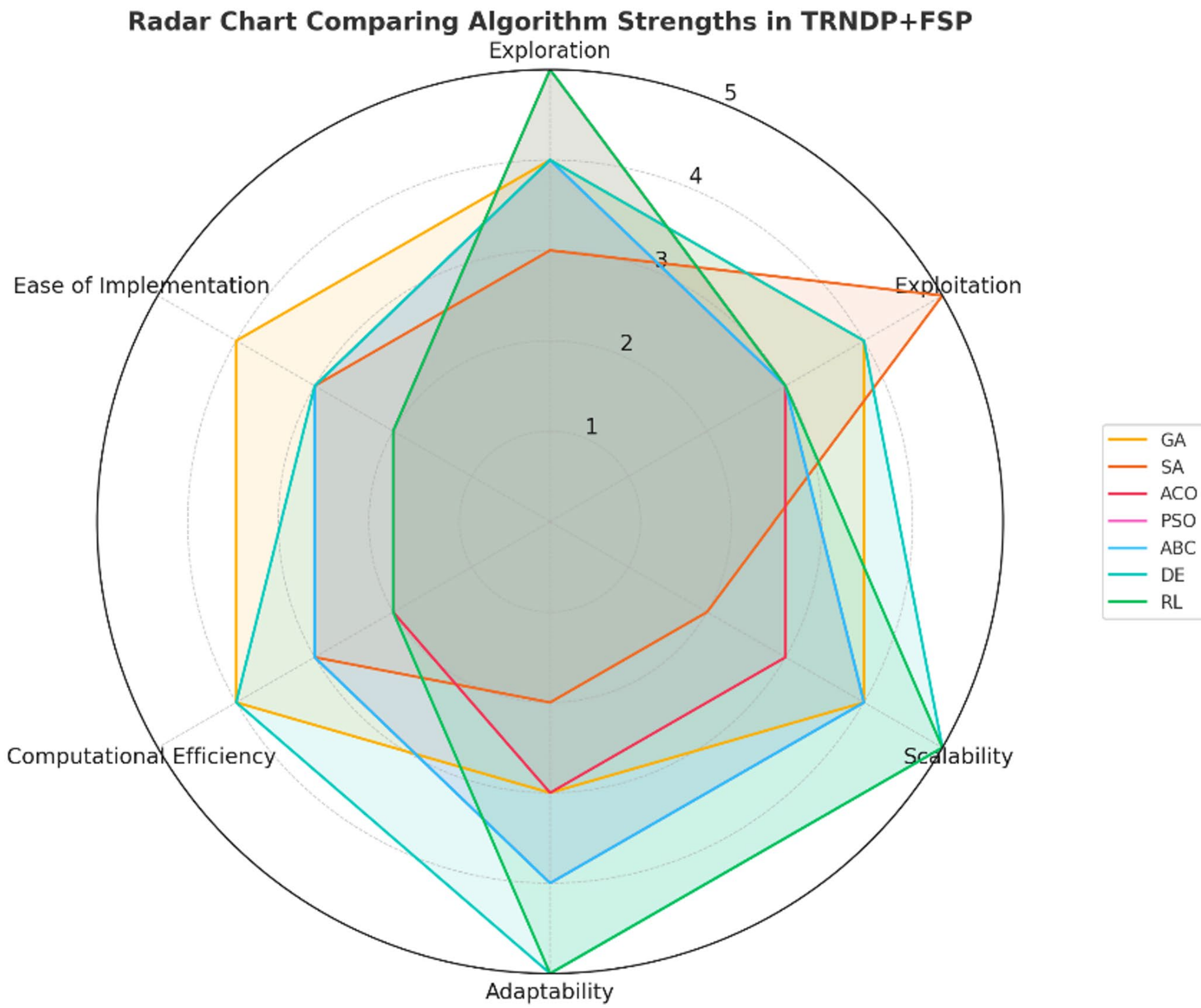

Fig. 10 TNDP algorithms strength comparison

exploitation ability, reflecting its strength in local optimization, but its adaptability and scalability are more limited, which has led to its use primarily in smaller or constrained problem instances. ACO excels in exploration, fitting studies that seek global search performance, but suffers from relatively lower computational efficiency and ease of implementation, which restrict its widespread adoption. PSO and BCO show similar performance profiles with strong exploration and reasonable scalability, aligning with studies from the 2010s where these methods were applied for multi-objective TNDP tasks. Differential evolution (DE) is among the top-performing algorithms in scalability and adaptability, supporting its emerging role in recent years (especially in studies from the late 2010s onward) where it was used to manage multi-modal or intersecting networks. RL, although still underrepresented in literature, scores highest in adaptability and scalability, confirming its potential for real-time, data-driven transit design. However, it remains complex to implement and computationally demanding, which explains its limited use so far.

Although the integration of route network design and frequency setting has been extensively modeled in academic research, its practical implementation remains at an early stage. Most transit agencies continue to rely on sequential or heuristic planning processes due to institutional inertia, fragmented data environments, and the perceived risks of deploying complex optimization models in operational settings. Nevertheless, recent pilot projects in Europe and Asia demonstrate a gradual shift toward integrated decision-support platforms, often facilitated by the availability of open mobility data, smart card systems, and advanced simulation tools. These initiatives suggest that while integration is still primarily driven by academic research, it is beginning to find footholds in practice. The pace of adoption will depend less on methodological readiness, which is increasingly mature, and more on organizational willingness, cross-agency collaboration, and the development of governance frameworks that can sustain data-driven planning across multiple stakeholders [78, 158].

Resilience in transit network design can be conceptualized and modeled through several complementary approaches. Robust optimization focuses on ensuring performance under worst-case scenarios by safeguarding against demand uncertainty, link failures, or parameter mis-specification. Stochastic planning adopts a probabilistic perspective, explicitly modeling disruption likelihoods, demand variability, and recovery trajectories to provide solutions that balance efficiency with risk exposure. Meanwhile, redundancy-based design emphasizes structural continuity by embedding spare capacity, alternative routes, and multimodal backup options into the network. Each approach captures a distinct dimension of resilience: robust optimization ensures guaranteed service levels under uncertainty, stochastic planning reflects the probabilistic nature of real-world disruptions, and redundancy provides physical adaptability to shocks. Future research should not treat these as competing paradigms but rather as complementary layers of resilience. Hybrid models that integrate robust, stochastic, and redundancy-based strategies offer the most promising pathway toward transit systems that are not only efficient but also adaptive, inclusive, and reliable in the face of climate change, pandemics, and other systemic shocks [91],166.

Moving forward, equity in TNDP research should be operationalized through standardized and measurable indicators that enable comparability across studies and facilitate real-world benchmarking. Three categories of metrics stand out as particularly relevant. First, accessibility indices, such as cumulative opportunity measures (percentage of jobs, schools, or health facilities reachable within a fixed travel time) and gravity-based measures, can quantify geographic inclusivity. Second, distributional fairness metrics, including the Gini coefficient, Theil index, or other inequality measures applied to service supply or travel times, provide a rigorous way to assess whether resources are equitably distributed across populations and neighborhoods. Third, vulnerability-sensitive indicators can measure the proportion of low-income, elderly, or mobility-impaired residents within catchment areas of high-frequency or step-free transit services. Together, these metrics offer a transparent and replicable framework for embedding equity into TNDP models. Standardization across studies would not only enhance scholarly comparability but also provide policymakers with concrete tools to evaluate and monitor progress toward more inclusive transit systems.

A critical frontier for the next wave of TNDP research is the integration of real-time mobility data from ride-sharing and micromobility services into network design frameworks [7]. These data streams provide rich insights into first–last mile connectivity, demand volatility, and cross-modal substitution patterns, offering the potential to design transit networks that are more adaptive and multimodally integrated [6]. However, their incorporation also raises methodological challenges. The heterogeneity of data sources (e.g., ride-hailing platforms, bike-sharing systems, e-scooter providers), their volatility over short time horizons, and the lack of standardized data-sharing protocols complicate their direct use in large-scale optimization models. Integrating such data may therefore risk inflating model complexity and undermining scalability [131]. A promising direction is incremental integration, starting with aggregated mobility indicators that complement conventional transit data and progressively advancing toward real-time, AI-enhanced optimization frameworks capable of handling multimodal dynamics. In this way, micromobility and ride-sharing data can enhance TNDP without overwhelming its methodological tractability [126].

Conclusion

This study has provided the most comprehensive and temporally resolved review of the TNDP to date, spanning five decades and encompassing both the TRNDP and the FSP, as well as their integration. By systematically tracing methodological evolution, the review reveals critical transitions—from deterministic mathematical programming to the dominance of heuristics and metaheuristics in the 2000s and 2010s, and more recently, the gradual emergence of AI and ML approaches post-2020. Using percentage-based visual analytics, taxonomies, and a novel scalability–performance matrix, the study maps how methodological preferences have shifted over time, highlighting the plateauing of metaheuristics, the decline of classical optimization, and the limited adoption of AI to date. The scientific value added lies in four areas: (i) introducing a longitudinal perspective that captures five decades of TNDP evolution; (ii) developing a scalability–performance matrix to systematically evaluate solution methodologies against network size and complexity; (iii) explicitly analyzing TRNDP–FSP integration as an underexplored but critical research frontier; and (iv) setting forth a forward-looking agenda on equity, resilience, and AI-driven adaptability. These contributions advance both the theoretical and methodological foundations of TNDP research. From an applicability standpoint, the findings provide decision-support insights for city planners and transit authorities. The scalability–performance matrix offers a pathway to align methodological choices with network scale, data availability, and institutional capacity. The integration of TRNDP and FSP provides a blueprint for balancing operator efficiency with passenger accessibility. Moreover, by embedding equity and resilience into TNDP, this review highlights planning dimensions that are vital for post-pandemic, climate-sensitive, and socially diverse urban contexts. At the same time, this study has limitations. While over 170 studies were analyzed, the focus was primarily on methodological contributions, with less attention to political, institutional, and financial barriers that affect TNDP implementation. Furthermore, while AI, reinforcement learning, and digital twins are highlighted as promising, their adoption is constrained by data silos, interoperability challenges, and interpretability concerns. Future research should therefore prioritize: (i) embedding standardized equity and resilience metrics—such as accessibility indices, Gini coefficients, and redundancy measures—into TNDP formulations; (ii) overcoming institutional barriers to data sharing to unlock the full potential of AI, RL, and digital twin applications; and (iii) developing hybrid, interpretable frameworks that combine classical optimization, metaheuristics, and AI for solutions that are both robust and operationally feasible.

Implications of this study extend to both academia and practice.

- *Academic contributions*: (a) providing the most temporally comprehensive synthesis of TNDP research; (b) introducing novel analytical frameworks including taxonomies, heatmaps, and the scalability–performance matrix; and (c) identifying underexplored frontiers in equity, resilience, and AI-based transit planning.
- *Practical contributions*: (a) offering a decision-support framework to guide agencies in method selection; (b) demonstrating how integrated TRNDP–FSP models can enhance coordinated planning; and (c) highlighting the urgency of embedding inclusivity and resilience into network design.

In sum, this review not only consolidates half a century of TNDP scholarship but also sets a strategic roadmap for the next wave of research and practice. Bridging classical optimization with modern, data-rich paradigms is essential for creating transit networks that are efficient, equitable, resilient, and technologically adaptive, capable of meeting the challenges of next-generation urban mobility.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval All authors have read and approved the final manuscript and agree to its submission to the journal.

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